C Y B E R S T E R I N T E R N S H I P P R O G R A M

WEEK 2 REPORT

Network Enumeration & Service Vulnerability Discovery

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I N S T I T U T I O N
CYBERSTER

R E P O R T P E R I O D
Week 2

S U B M I S S I O N D A T E
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1. Executive Summary

Week 2 of the Cyberster internship focused on transitioning from passive reconnaissance to active network scanning and enumeration. Building upon the foundational OSINT techniques learned in Week 1, this week's exercises involved comprehensive network scanning, service fingerprinting, vulnerability discovery, and stealth techniques.

The primary targets for this week's exercises included Metasploitable 2 and various lab-configured systems. Through systematic execution of four major tasks, this report documents the successful identification of open ports, service versions, potential vulnerabilities, and the application of evasion techniques to bypass basic security controls.

Key Finding: The transition from passive to active reconnaissance significantly increases attack surface visibility but also introduces detection risks. Understanding the balance between thoroughness and stealth is critical for professional security assessments.

2. Learning Objectives

Upon completion of Week 2, the following objectives were achieved:

- Understand the mechanics of the TCP 3-way handshake and connection establishment
- Differentiate between TCP Connect scans (-sT) and SYN Stealth scans (-sS) in terms of speed, visibility, and detection risk
- Perform UDP scanning and understand its unique challenges due to the stateless nature of UDP
- Apply timing templates to control scan speed and understand their impact on reliability and detection
- Identify exact service versions using Nmap's version detection (-sV)
- Perform manual banner grabbing using Netcat and Telnet
- Enumerate SMB shares and users using Enum4linux
- Utilize the Nmap Scripting Engine (NSE) for automated vulnerability discovery
- Map NSE output to specific CVE IDs
- Apply stealth and firewall evasion techniques including fragmentation, decoys, and source port manipulation

3. Environment Setup and Tools

Operating System: Kali Linux 2026.2 (Debian-based, specialized for penetration testing)

Virtualization: VirtualBox 7.0 with Linux kernel modules

VM Configuration: 4 cores CPU | 8 GB RAM | 60 GB Disk | Bridged Network

Target Systems: Metasploitable 2 (10.0.2.3) and other lab-configured targets

Tool	Version	Purpose
Nmap	7.99+	Network scanning, service fingerprinting, NSE scripts
Netcat (nc)	1.217+	Manual banner grabbing
Telnet	0.17+	Manual banner grabbing
Enum4linux	0.9.1+	SMB enumeration
Metasploitable 2	N/A	Vulnerable target system

4. Task 1: Advanced Network Scanning with Nmap

Objective

To transition from broad reconnaissance to targeted network scanning, identifying open ports and fingerprinting services while avoiding basic security triggers.

Theory and Background

Network scanning is a critical phase in security assessment that involves sending packets to target systems to determine which ports are open and what services are running. Nmap (Network Mapper) is the industry-standard tool for this purpose.

The TCP 3-Way Handshake:

- Client → SYN: Client requests to open a connection
- Server → SYN-ACK: Server acknowledges and offers to synchronize
- Client → ACK: Client confirms; connection established

Why it matters for Nmap: -sT (TCP Connect) completes the full 3-way handshake, while -sS (SYN Stealth) sends only the SYN and never completes the handshake, so the connection remains half-open and is less likely to be logged.

Task 1 Screenshots

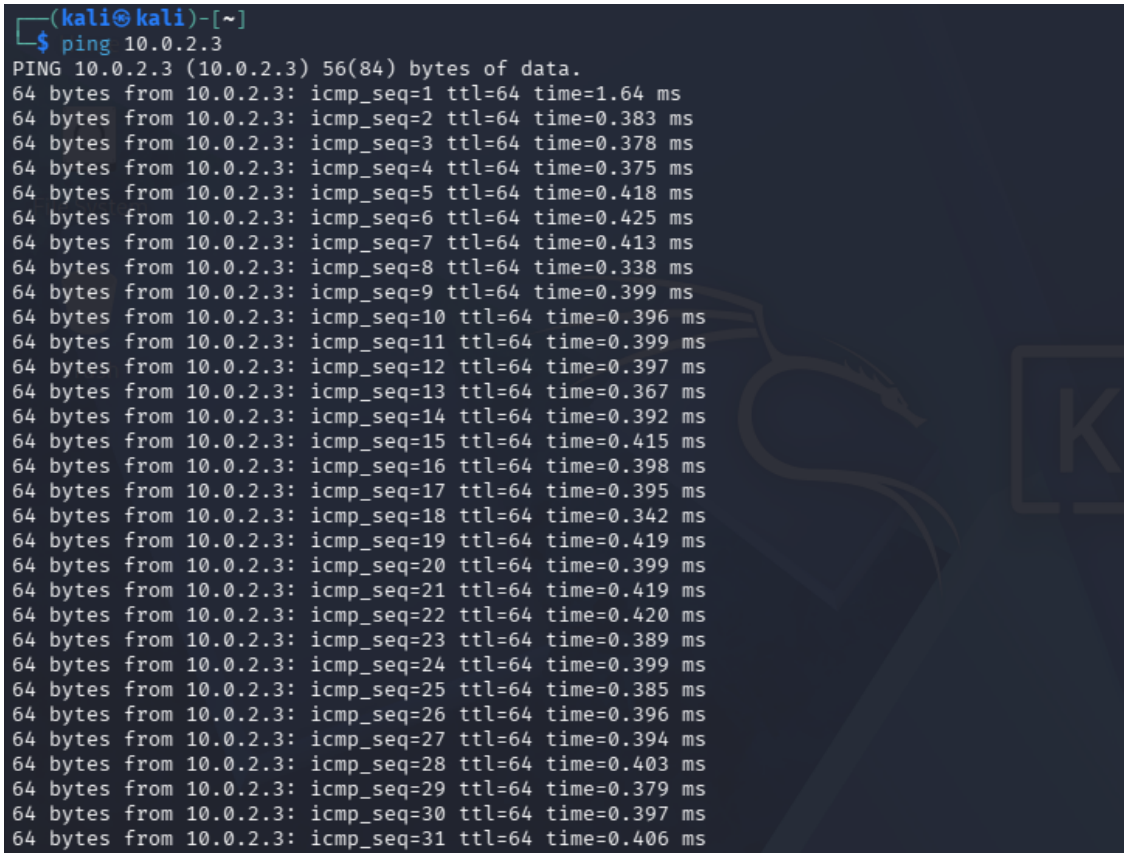


Figure 1.1

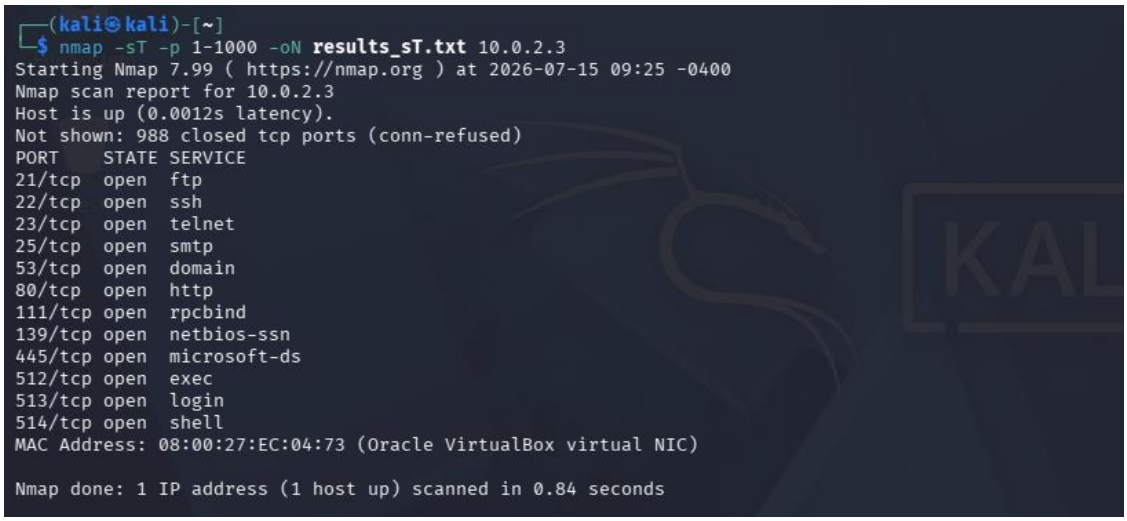


Figure 1.2

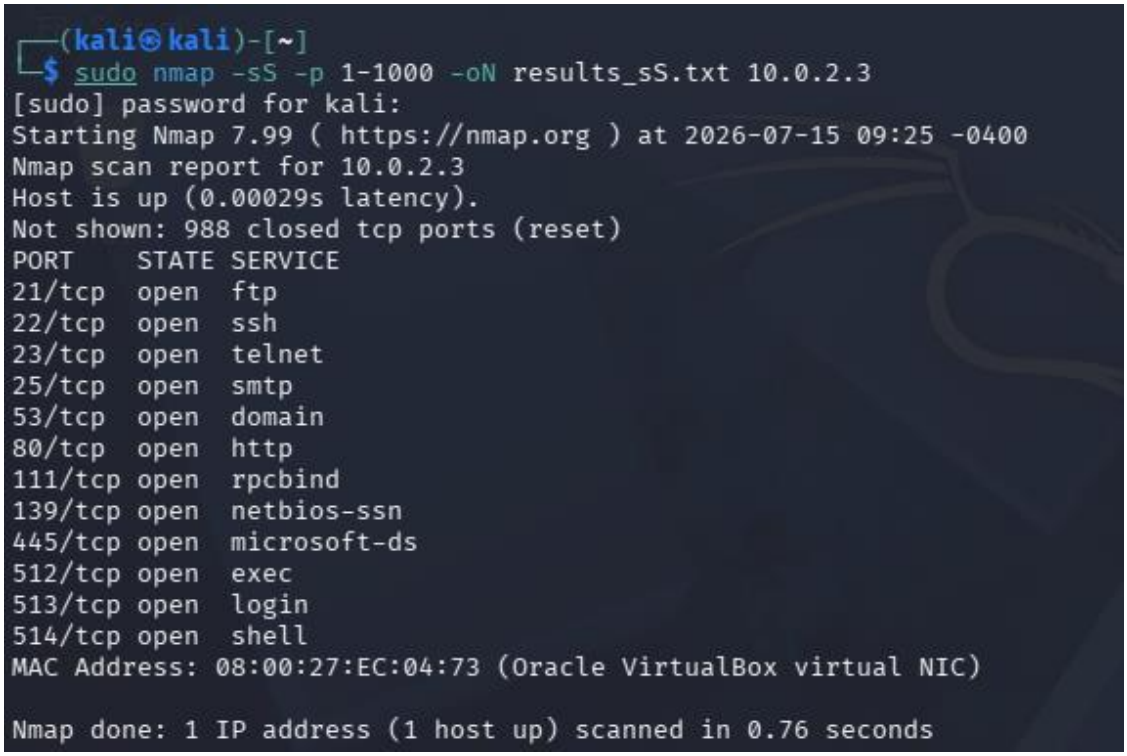


Figure 1.3

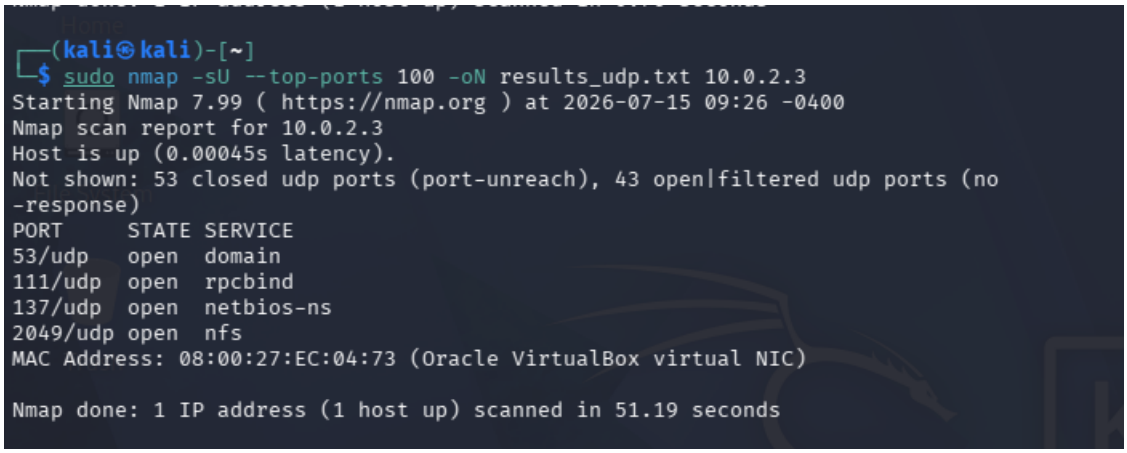


Figure 1.4

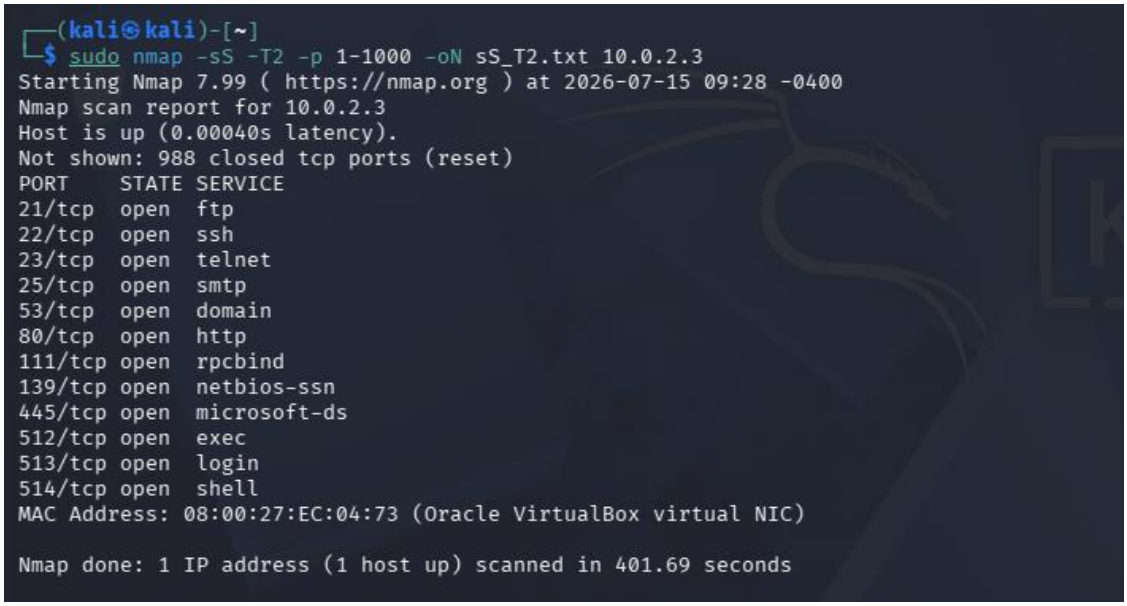


Figure 1.5

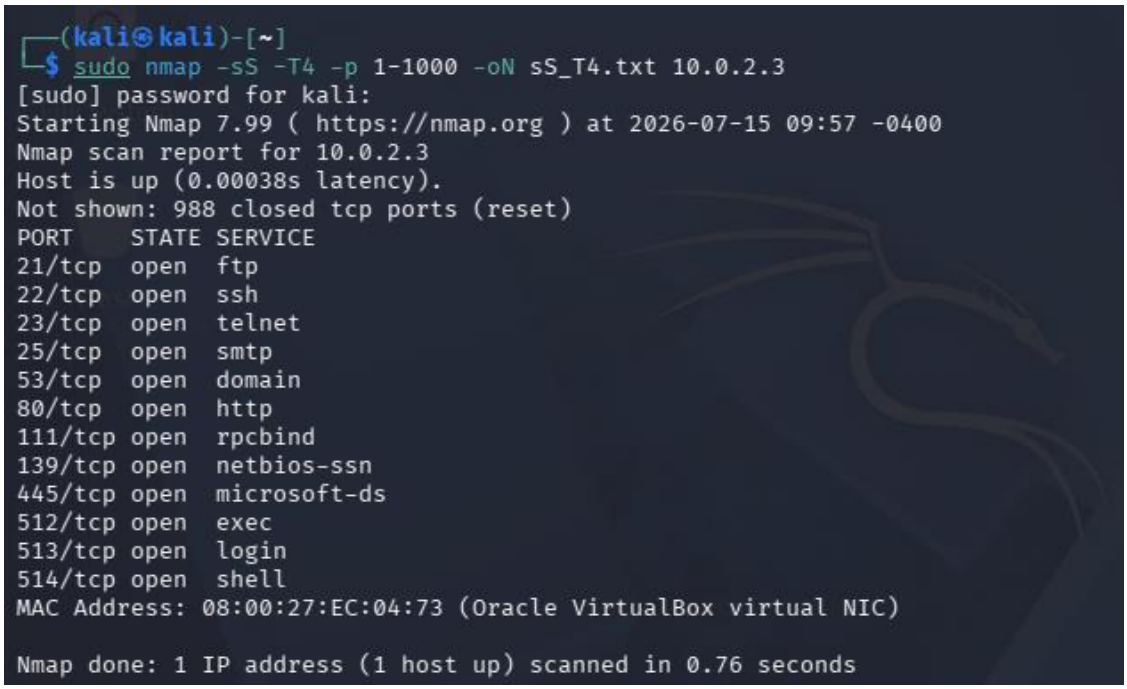


Figure 1.6

Task 1, Part 1: TCP Connect Scan (-sT)

Figure 1.1 — TCP Connect scan (-sT) against 10.0.2.3, ports 1–1000

Analysis:

- Open Ports Identified: 21 (FTP), 22 (SSH), 23 (Telnet), 25 (SMTP), 53 (DNS), 80 (HTTP), 111 (RPC), 139/445 (SMB), 512–514 (r-services)
- Scan Duration: Completed very quickly (0.84 seconds) on the local VM
- Visibility: This scan is more likely to be logged by target systems, as it establishes complete connections

Task 1, Part 2: SYN Stealth Scan (-sS)

Figure 1.2 — SYN Stealth scan (-sS) against 10.0.2.3, ports 1–1000

Comparison: -sT vs -sS

Aspect	TCP Connect (-sT)	SYN Stealth (-sS)
Handshake	Complete (SYN, SYN-ACK, ACK)	Half-open (SYN, SYN-ACK, RST)
Visibility	More likely logged	Less likely logged
Requires Root	No	Yes (sudo)

Task 1, Part 3: UDP Scanning (-sU)

Figure 1.3 — UDP scan of the top 100 ports against 10.0.2.3

Task 1, Part 4: Timing Templates (-T2 vs -T4)

Figure 1.4 — SYN scan run with -T2 (polite) and -T4 (aggressive) timing templates

Interpretation: The T2 polite scan took over 400 seconds, significantly reducing detection risk. The T4 aggressive scan took under 1 second but was considerably noisier on the network.

5. Task 2: Service Fingerprinting & Banner Grabbing

Task 2 Screenshots

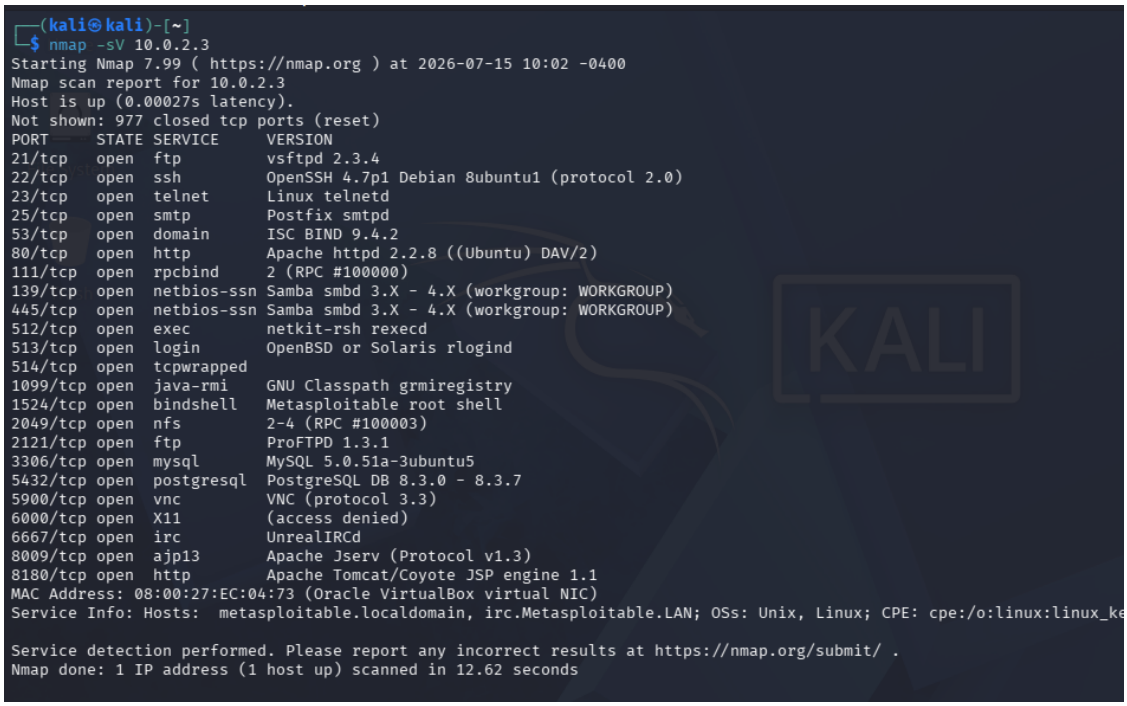


Figure 2.1



Figure 2.2

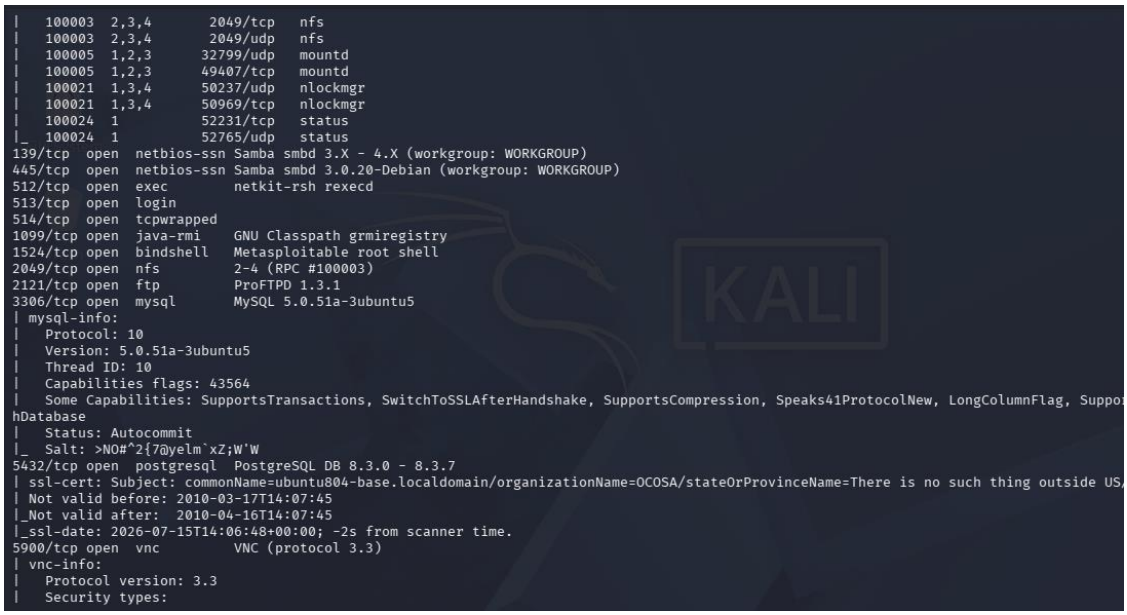


Figure 2.3

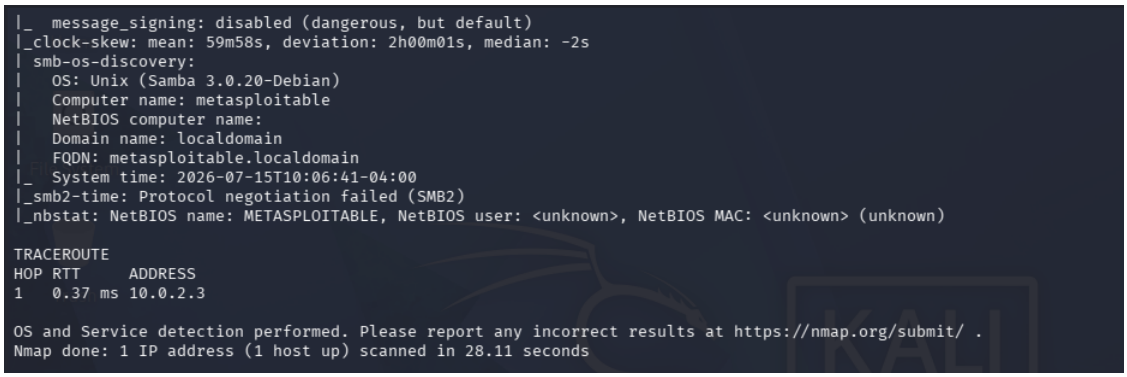


Figure 2.4



Figure 2.5

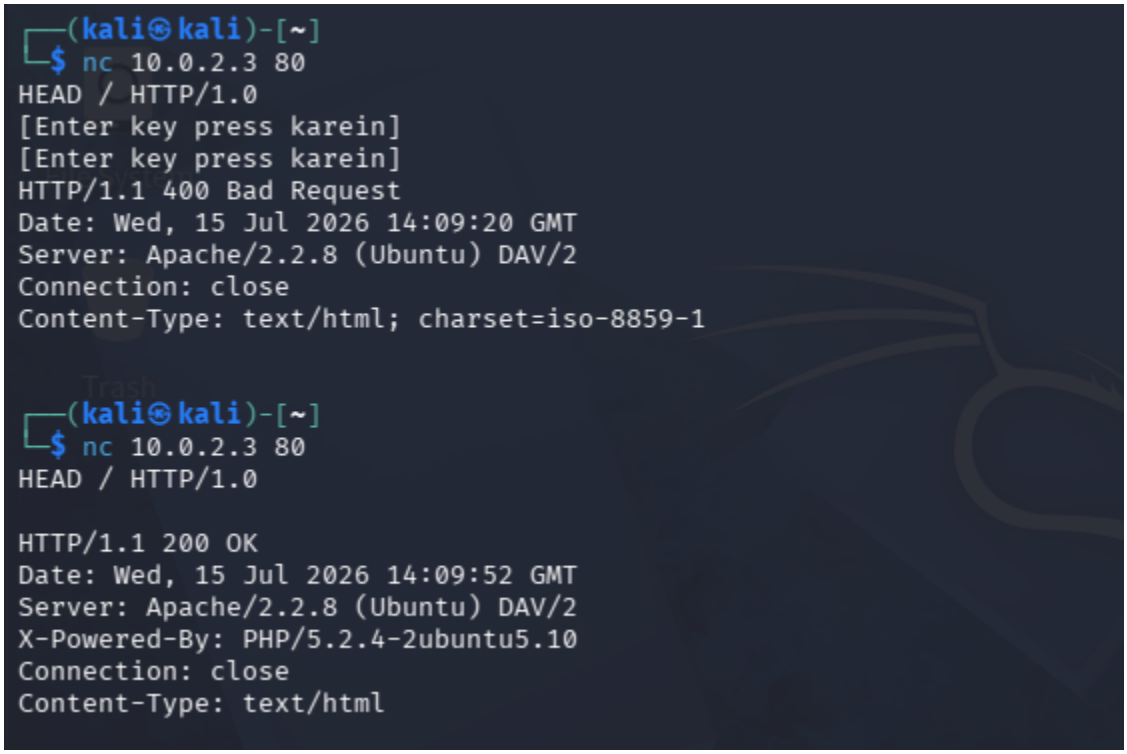


Figure 2.6

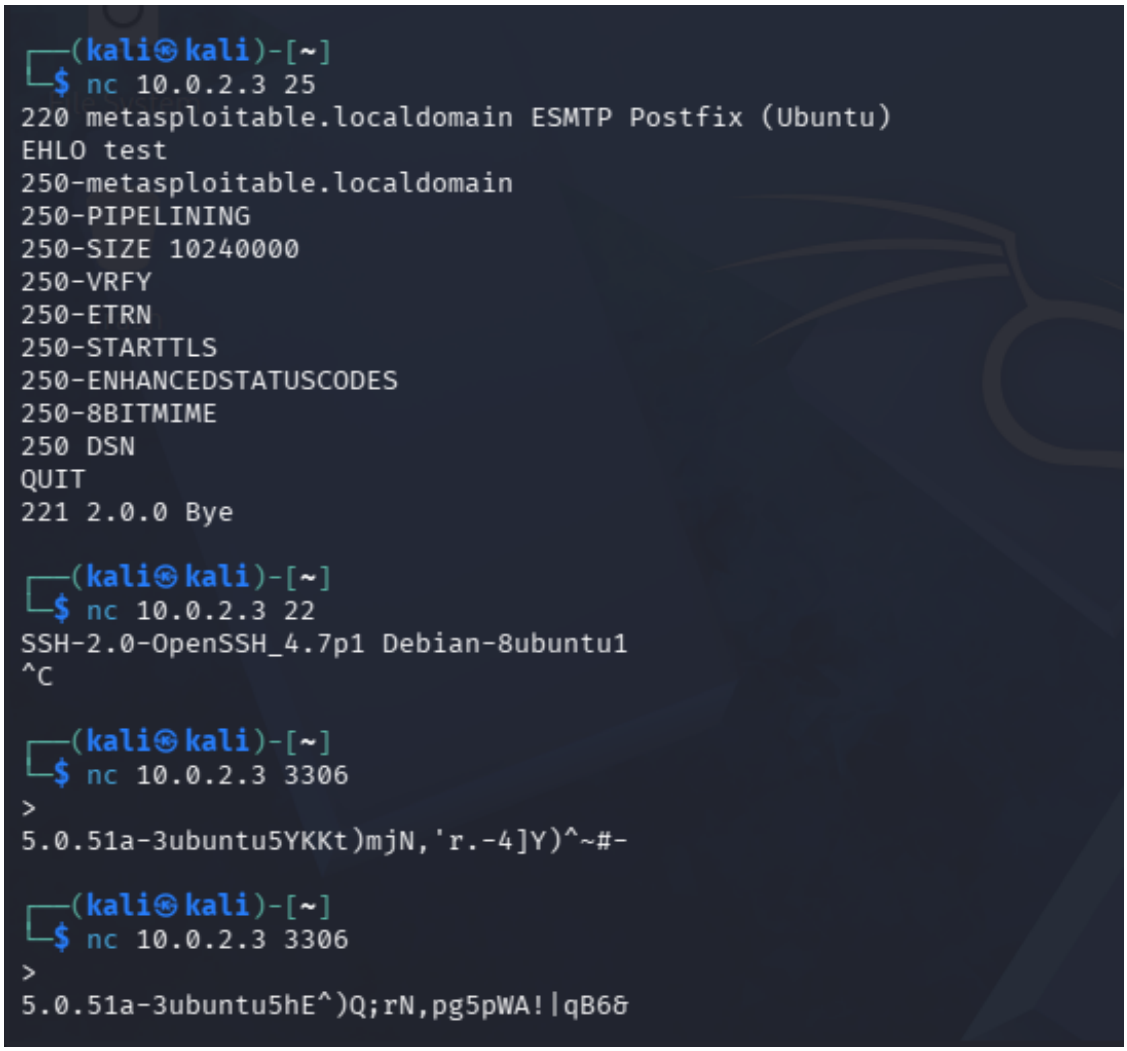

┌──(kali㉿kali)-[~]
└─$ nc 10.0.2.3 25
220 metasploitable.localdomain ESMTP Postfix (Ubuntu)
EHLO test
250-metasploitable.localdomain
250-PIPELINING
250-SIZE 10240000
250-VRFY
250-ETRN
250-STARTTLS
250-ENHANCEDSTATUSCODES
250-8BITMIME
250 DSN
QUIT
221 2.0.0 Bye
┌──(kali㉿kali)-[~]
└─$ nc 10.0.2.3 22
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
^C
┌──(kali㉿kali)-[~]
└─$ nc 10.0.2.3 3306
>
5.0.51a-3ubuntu5YKKt)mjN,'r.-4]Y)^~#-
┌──(kali㉿kali)-[~]
└─$ nc 10.0.2.3 3306
>
5.0.51a-3ubuntu5hE^)Q;rN,pg5pWA!|qB6&

Figure 2.7

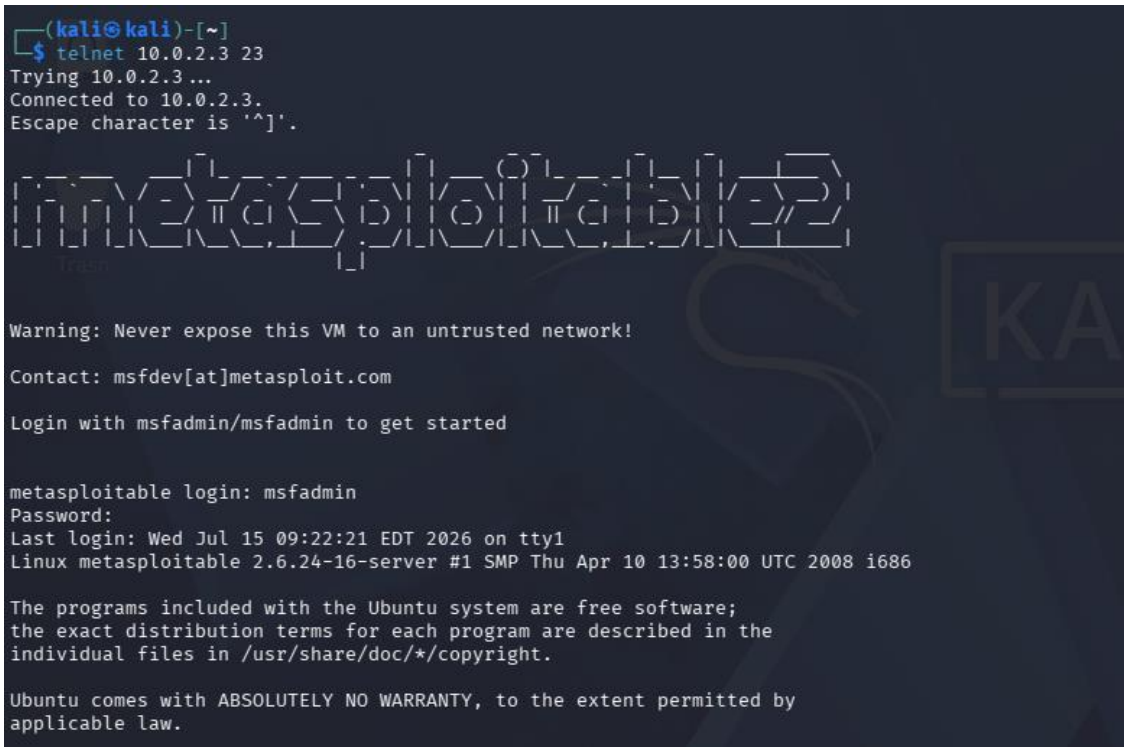


Figure 2.8

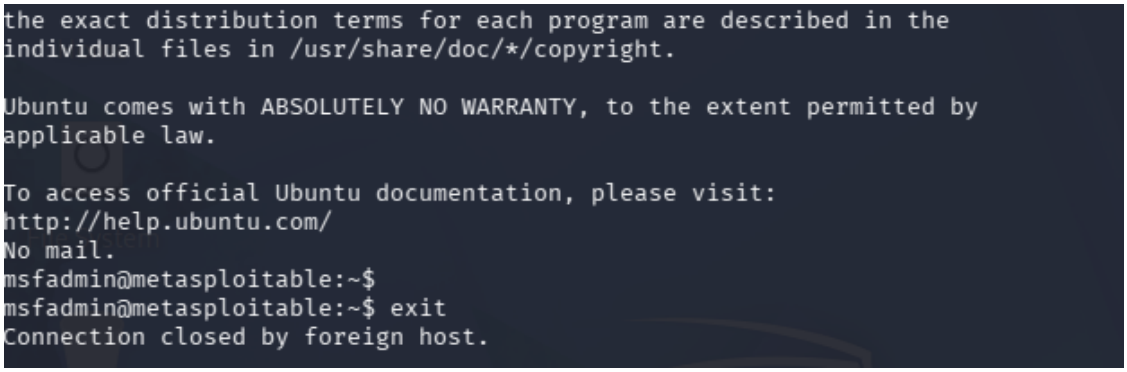


Figure 2.9

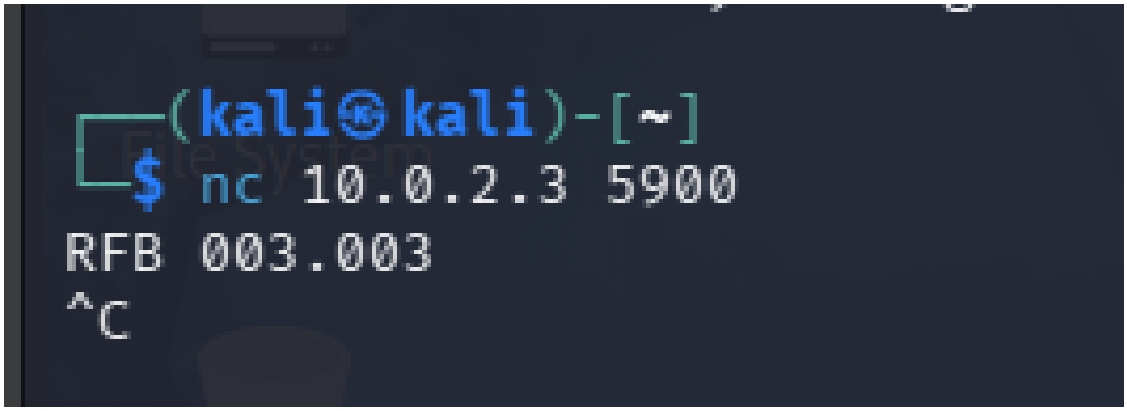


Figure 2.10

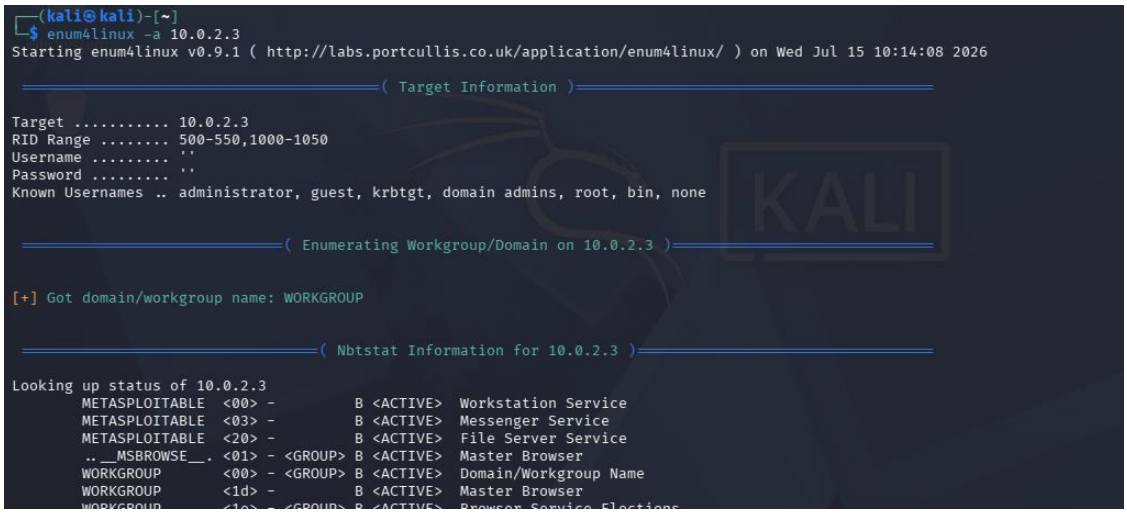


Figure 2.11

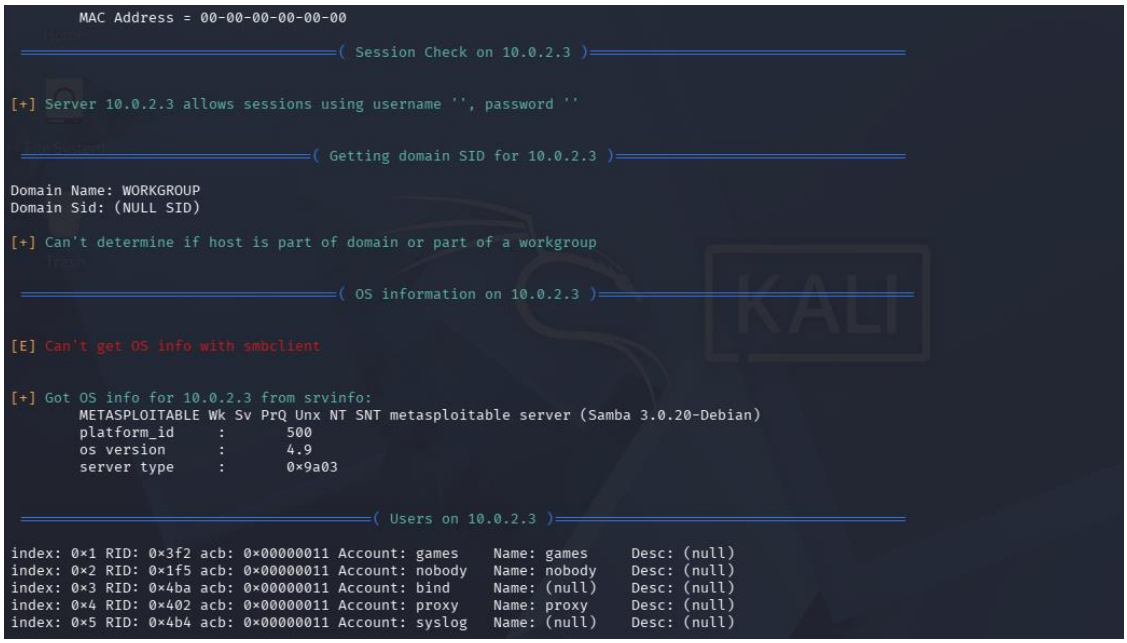


Figure 2.12

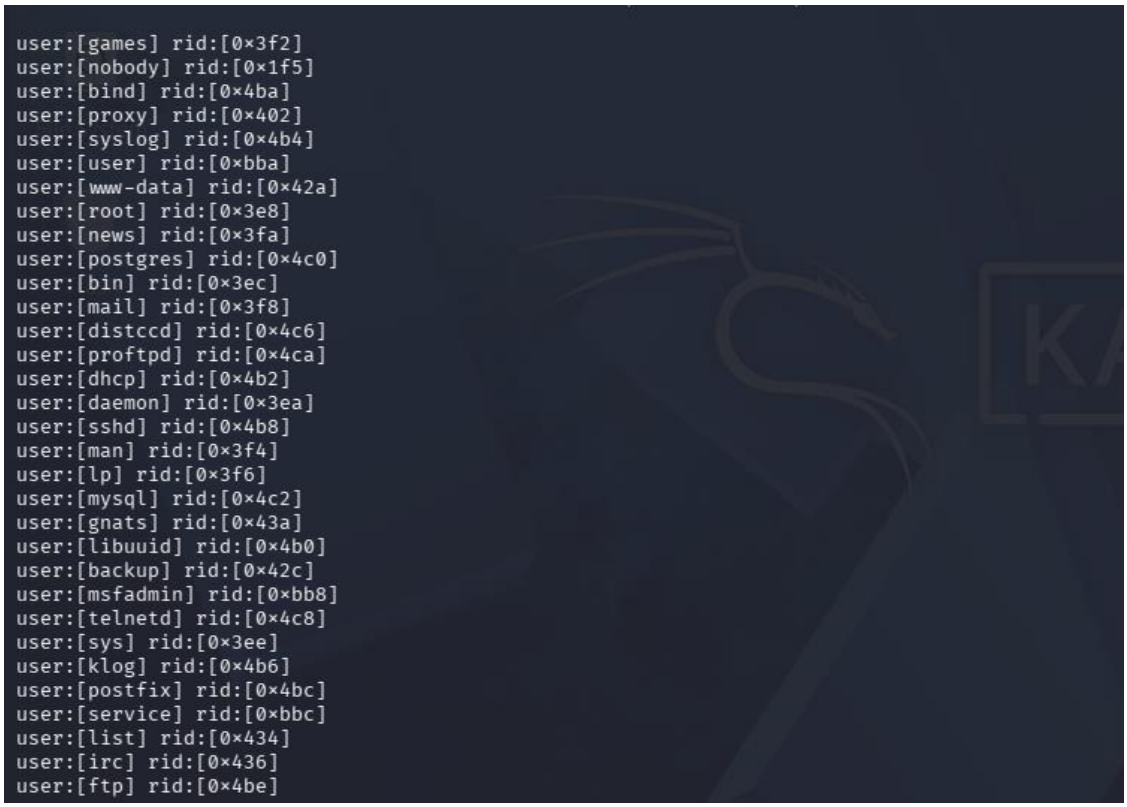


Figure 2.13

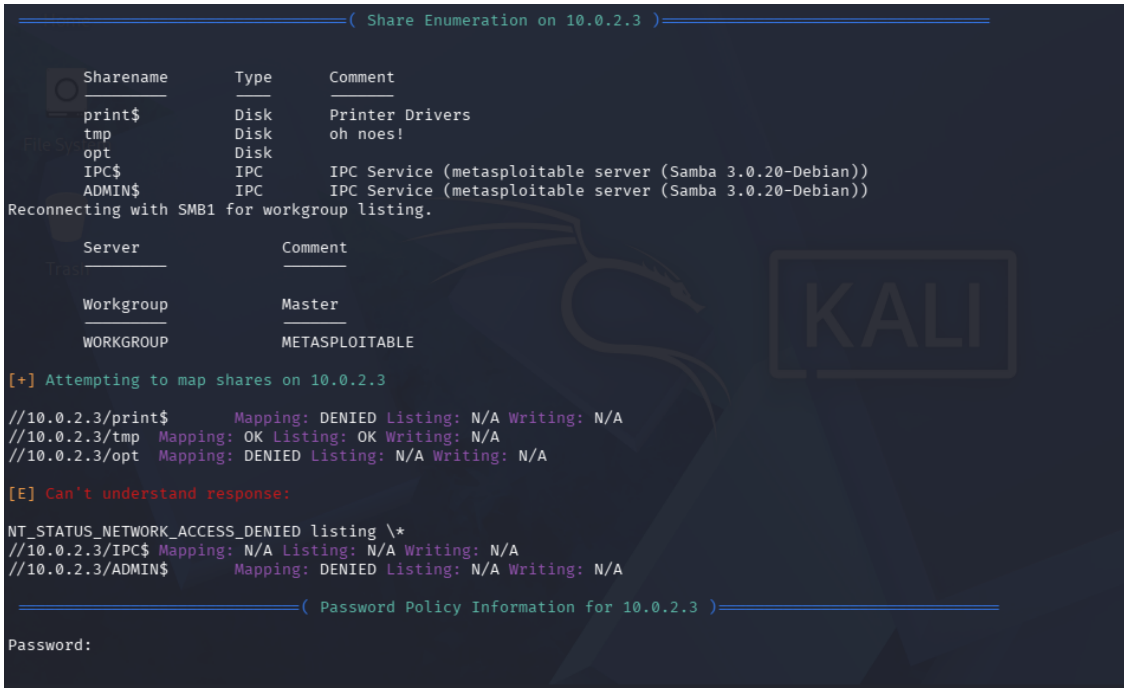


Figure 2.14

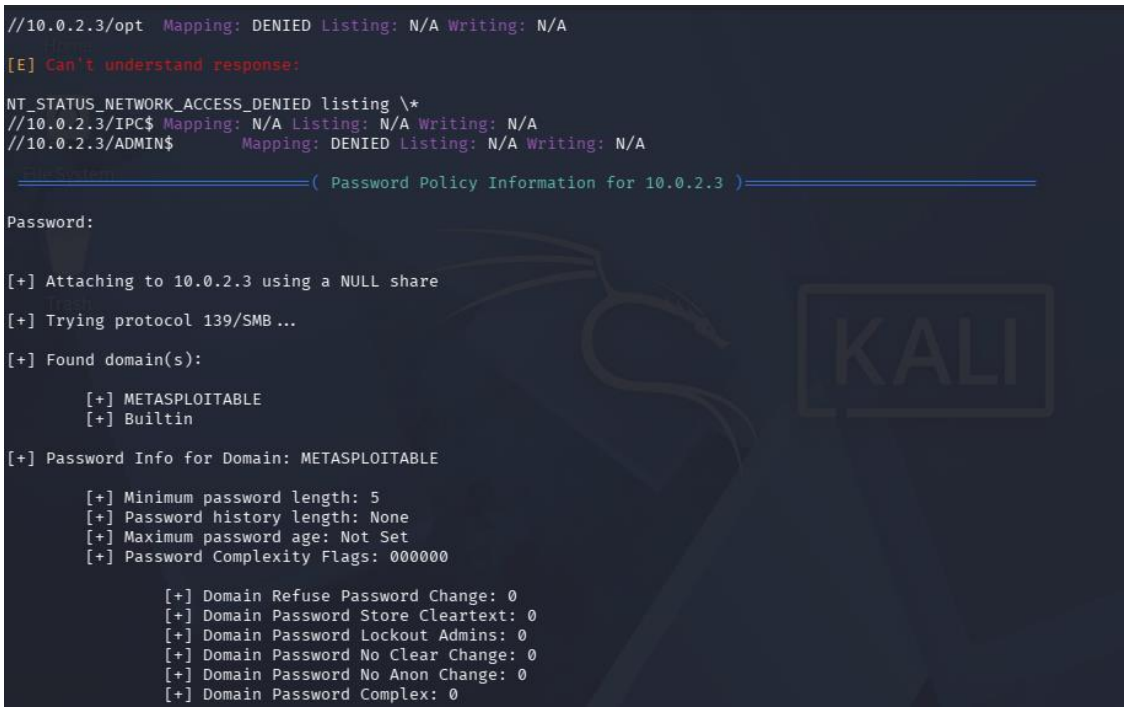


Figure 2.15

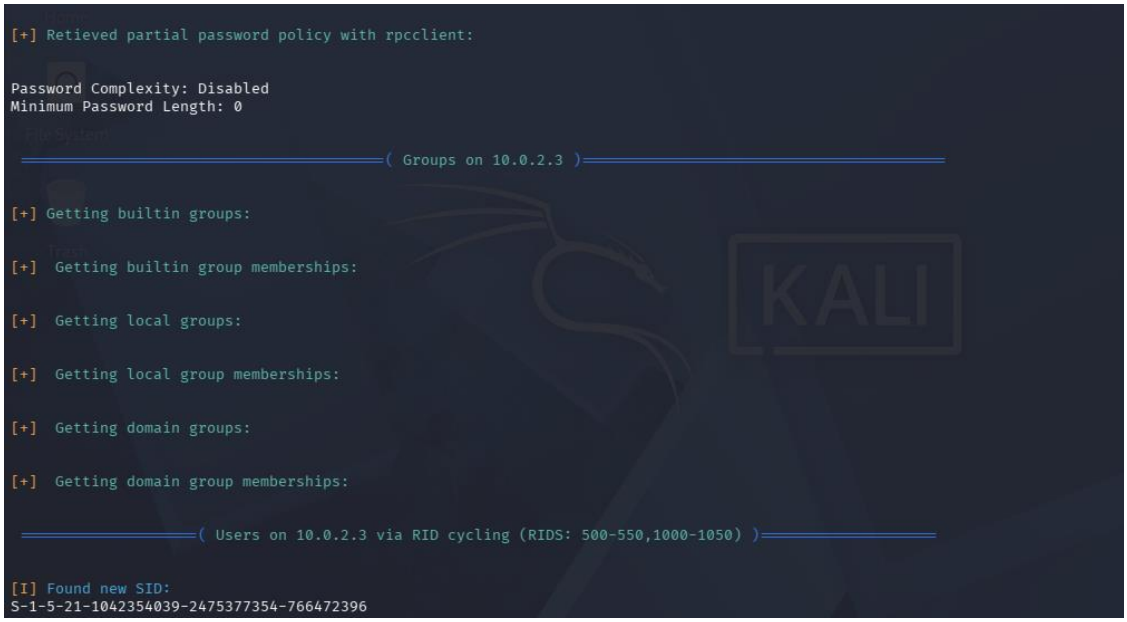


Figure 2.16

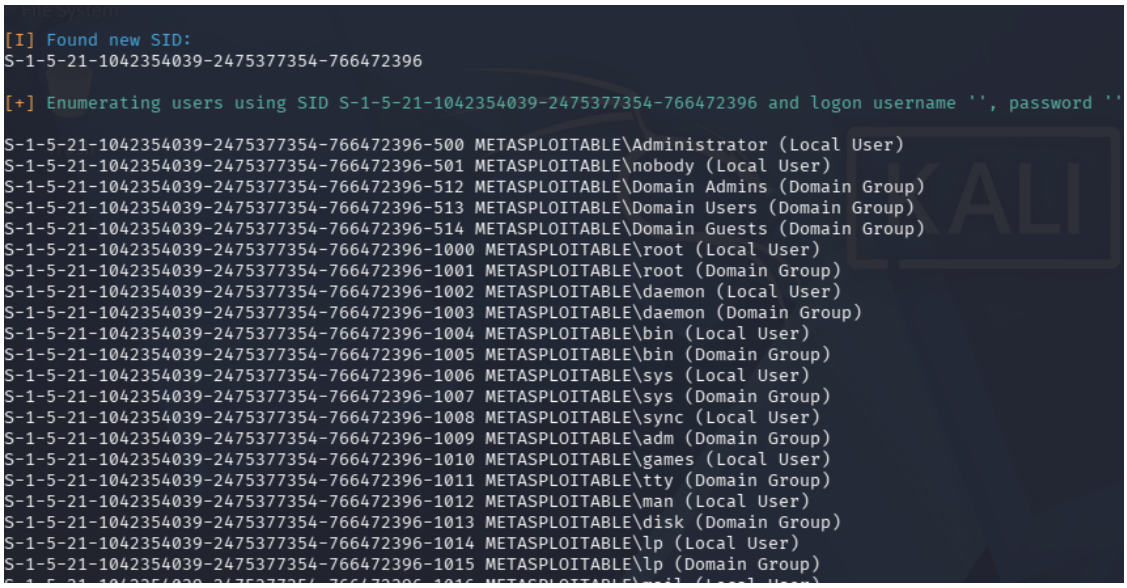


Figure 2.17

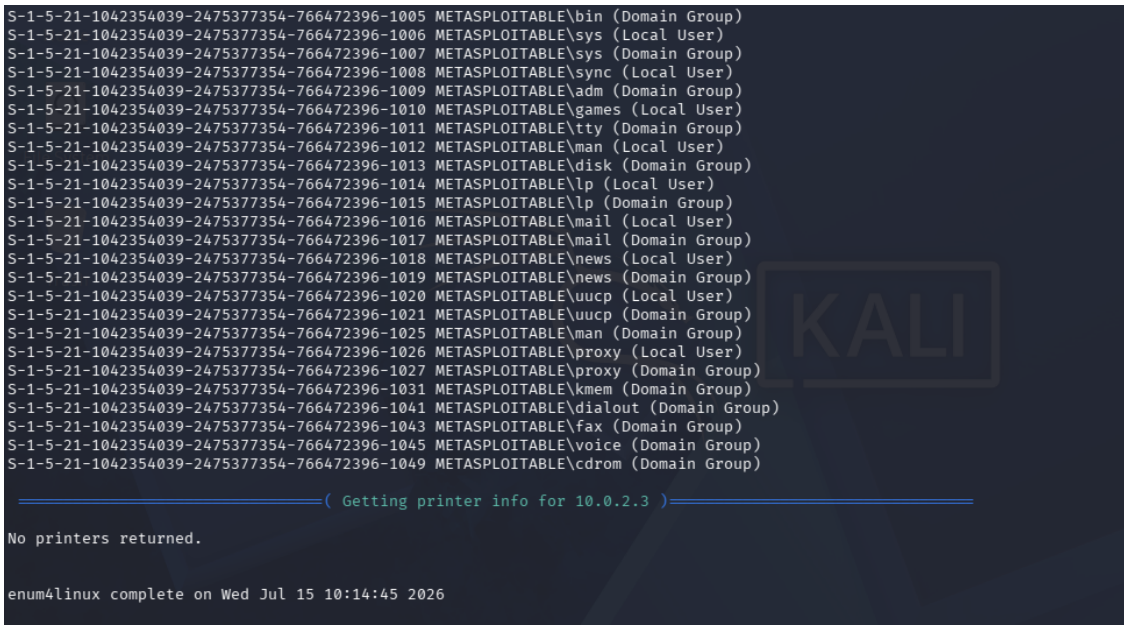


Figure 2.18

Task 2, Part 1: Service Version Detection (-sV) & Aggressive Scan (-A)

Figure 2.1 — Nmap version detection (-sV) revealing exact service versions

Figure 2.2 — Aggressive scan (-A) with OS detection, script scanning, and traceroute

Task 2, Part 2: Manual Banner Grabbing

Figure 2.3 — Manual banner grabbing with Netcat (FTP, HTTP) and Telnet

Task 2, Part 3: SMB Enumeration (Port 445)

Figure 2.4 — SMB enumeration using enum4linux -a

6. Task 3: Leveraging the Nmap Scripting Engine (NSE)

Task 3 Screenshots

```
┌──(kali㉿kali)-[~]
└─$ nmap -p 80 --script http-enum 10.0.2.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 10:24 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00042s latency).

PORT   STATE SERVICE
80/tcp open  http
| http-enum:
|   /tikiwiki/: Tikiwiki
|   /test/: Test page
|   /phpinfo.php: Possible information file
|   /phpMyAdmin/: phpMyAdmin
|   /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'
|   /icons/: Potentially interesting folder w/ directory listing
|_  /index/: Potentially interesting folder
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds

┌──(kali㉿kali)-[~]
└─$ nmap -p 443 --script ssl-cert 10.0.2.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 10:24 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00040s latency).

PORT    STATE  SERVICE
443/tcp closed https
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.78 seconds
```

Figure 3.1

```
┌──(kali㉿kali)-[~]
└─$ nmap --script discovery 10.0.2.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 10:24 -0400
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Script Pre-Scan
NSE Timing: About 42.55% done; ETC: 10:24 (0:00:00 remaining)
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Script Pre-Scan
NSE Timing: About 40.82% done; ETC: 10:24 (0:00:01 remaining)
No profinet devices in the subnet
Pre-scan script results:
| ipv6-multicast-mld-list:
|   fe80::a00:27ff:feec:473:
|     device: eth0
|     mac: 08:00:27:ec:04:73
|     multicast_ips:
|_      ff02::1:ffec:473          (NDP Solicited-node)
| targets-ipv6-multicast-echo:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|_  Use --script-args=newtargets to add the results as targets
| targets-asn:
|_  targets-asn.asn is a mandatory parameter
| targets-ipv6-multicast-mld:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|
|_  Use --script-args=newtargets to add the results as targets
| targets-sniffer: Sniffed 8 address(es).
| 10.0.2.1
| fe80::def5:75c3:4069:c6a4
| fe80::a00:27ff:feec:473
| fe80::a00:27ff:fe5a:87bc
| 224.0.0.1
| 224.0.0.13
| 224.0.23.12
|_::
```

Figure 3.2

```
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|_  Use --script-args=newtargets to add the results as targets
| targets-asn:
|_  targets-asn.asn is a mandatory parameter
| targets-ipv6-multicast-mld:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|
|_  Use --script-args=newtargets to add the results as targets
| targets-sniffer: Sniffed 8 address(es).
| 10.0.2.1
| fe80::def5:75c3:4069:c6a4
| fe80::a00:27ff:feec:473
| fe80::a00:27ff:fe5a:87bc
| 224.0.0.1
| 224.0.0.13
| 224.0.23.12
|_::
|_http-robtex-shared-ns: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
|_hostmap-robtex: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
| targets-ipv6-multicast-invalid-dst:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|_  Use --script-args=newtargets to add the results as targets
| targets-ipv6-multicast-slaac:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|_  Use --script-args=newtargets to add the results as targets
```

Figure 3.3

```
┌──(kali㉿kali)-[~]
└─$ nmap --script safe 10.0.2.3
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 10:27 -0400
No profinet devices in the subnet
Pre-scan script results:
| targets-sniffer: Sniffed 14 address(es).
| 224.0.0.1
| 10.0.2.1
| 239.255.255.250
| 239.255.255.253
| 10.0.2.2
| 224.0.0.251
| 224.0.0.9
| 10.0.2.3
| fe80::def5:75c3:4069:c6a4
| fe80::a00:27ff:feec:473
| 0.0.0.0
| 224.0.0.13
| 192.168.0.1
|_224.0.23.12
|_hostmap-robtex: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
| broadcast-dhcp-discover:
|   Response 1 of 1:
|     Interface: eth0
|     IP Offered: 10.0.2.5
|     Server Identifier: 10.0.2.2
|     Subnet Mask: 255.255.255.0
|     Router: 10.0.2.1
|_    Domain Name Server: 192.168.0.1
| broadcast-netbios-master-browser:
| ip        server          domain
|_10.0.2.3  METASPLOITABLE  WORKGROUP
| broadcast-listener:
|   udp
|       DHCP
```

Figure 3.4

```
|     Router: 10.0.2.1
|_    Domain Name Server: 192.168.0.1
| broadcast-netbios-master-browser:
| ip        server          domain
|_10.0.2.3  METASPLOITABLE  WORKGROUP
| broadcast-listener:
|   udp
|       DHCP
|         srv ip    cli ip    mask           gw        dns          vendor
|_        10.0.2.2  10.0.2.5  255.255.255.0  10.0.2.1  192.168.0.1  -
|_http-robtex-shared-ns: *TEMPORARILY DISABLED* due to changes in Robtex's API. See https://www.robtex.com/api/
| targets-asn:
|_  targets-asn.asn is a mandatory parameter
| targets-ipv6-multicast-mld:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|
|_  Use --script-args=newtargets to add the results as targets
| eap-info:
| Available authentication methods with identity="anonymous" on interface eth0
|   unknown  EAP-TTLS
|   unknown  EAP-TLS
|   unknown  EAP-MSCHAP-V2
|_  unknown  PEAP
Bug in broadcast-upnp-info: no string output.
| targets-ipv6-multicast-invalid-dst:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
|_  Use --script-args=newtargets to add the results as targets
| targets-ipv6-multicast-echo:
|   IP: fe80::a00:27ff:feec:473  MAC: 08:00:27:ec:04:73  IFACE: eth0
```

Figure 3.5

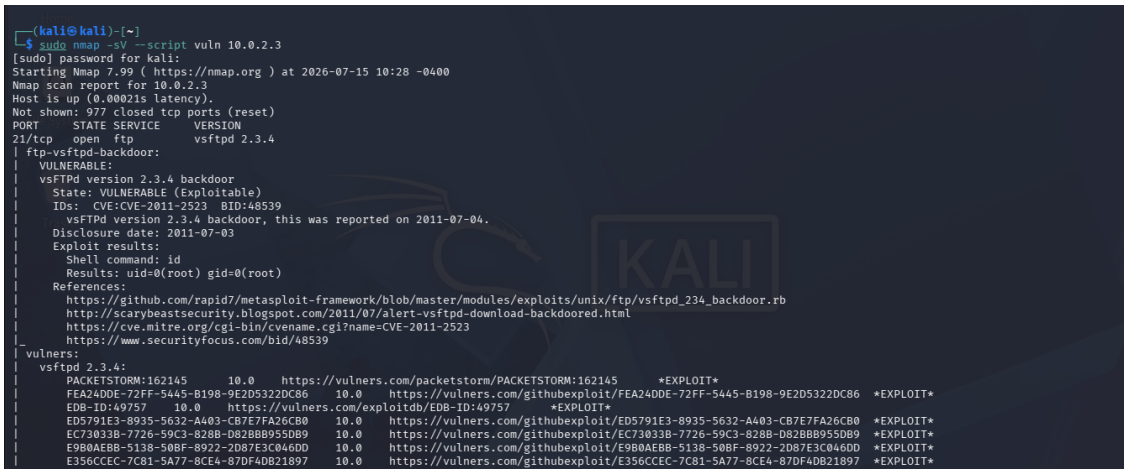


Figure 3.6

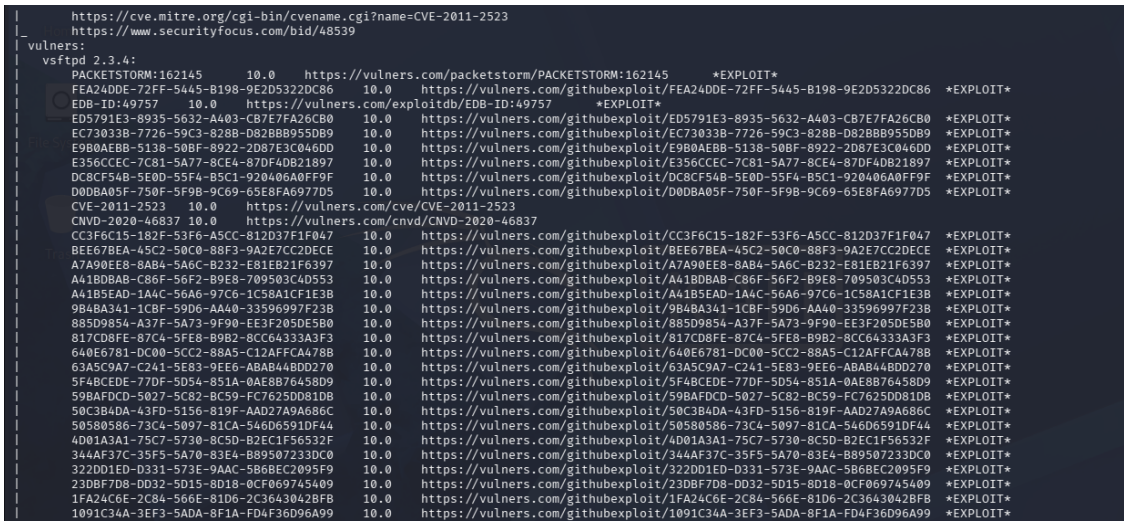


Figure 3.7

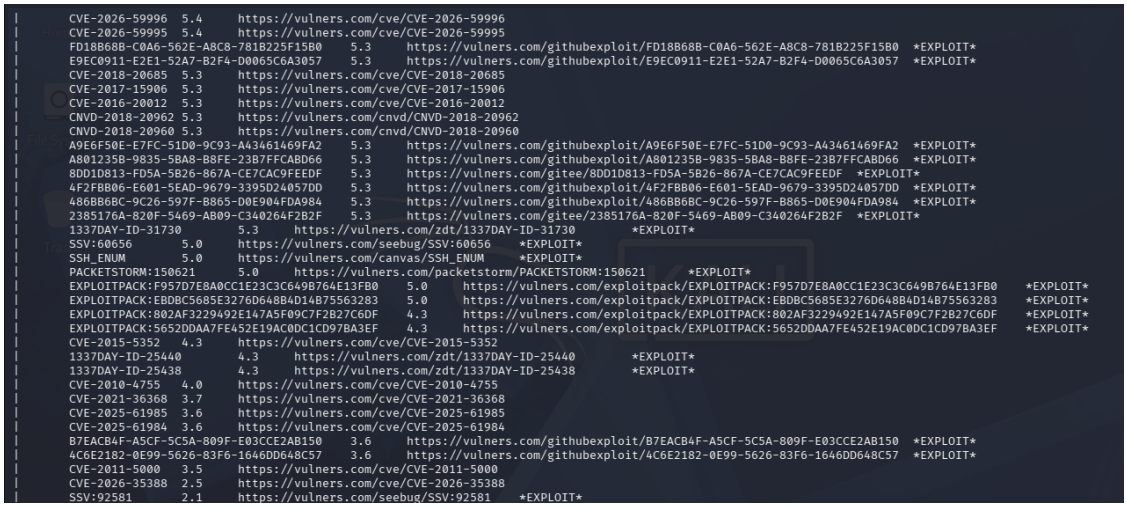


Figure 3.8

```
|_      1337DAY-ID-25391       0.0     https://vulners.com/zdt/1337DAY-ID-25391       *EXPLOIT*
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
| smtp-vuln-cve2010-4344:
|_  The SMTP server is not Exim: NOT VULNERABLE
| ssl-dh-params:
|   VULNERABLE:
|   Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
|     State: VULNERABLE
|       Transport Layer Security (TLS) services that use anonymous
|       Diffie-Hellman key exchange only provide protection against passive
|       eavesdropping, and are vulnerable to active man-in-the-middle attacks
|       which could completely compromise the confidentiality and integrity
|       of any data exchanged over the resulting session.
|     Check results:
|       ANONYMOUS DH GROUP 1
|             Cipher Suite: TLS_DH_anon_EXPORT_WITH_RC4_40_MD5
|             Modulus Type: Safe prime
|             Modulus Source: Unknown/Custom-generated
|             Modulus Length: 512
|             Generator Length: 8
|             Public Key Length: 512
|     References:
|       https://www.ietf.org/rfc/rfc2246.txt
|
|   Transport Layer Security (TLS) Protocol DHE_EXPORT Ciphers Downgrade MitM (Logjam)
|     State: VULNERABLE
|     IDs:  CVE:CVE-2015-4000  BID:74733
|       The Transport Layer Security (TLS) protocol contains a flaw that is
|       triggered when handling Diffie-Hellman key exchanges defined with
|       the DHE_EXPORT cipher. This may allow a man-in-the-middle attacker
|       to downgrade the security of a TLS session to 512-bit export-grade
|       cryptography, which is significantly weaker, allowing the attacker
|       to more easily break the encryption and monitor or tamper with
|       the encrypted stream.
```

Figure 3.9

```
|     IDs:  CVE:CVE-2015-4000  BID:74733
|       The Transport Layer Security (TLS) protocol contains a flaw that is
|       triggered when handling Diffie-Hellman key exchanges defined with
|       the DHE_EXPORT cipher. This may allow a man-in-the-middle attacker
|       to downgrade the security of a TLS session to 512-bit export-grade
|       cryptography, which is significantly weaker, allowing the attacker
|       to more easily break the encryption and monitor or tamper with
|       the encrypted stream.
|     Disclosure date: 2015-5-19
|     Check results:
|       EXPORT-GRADE DH GROUP 1
|             Cipher Suite: TLS_DHE_RSA_EXPORT_WITH_DES40_CBC_SHA
|             Modulus Type: Safe prime
|             Modulus Source: Unknown/Custom-generated
|             Modulus Length: 512
|             Generator Length: 8
|             Public Key Length: 512
|     References:
|       https://www.securityfocus.com/bid/74733
|       https://weakdh.org
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2015-4000
|
|   Diffie-Hellman Key Exchange Insufficient Group Strength
|     State: VULNERABLE
|       Transport Layer Security (TLS) services that use Diffie-Hellman groups
|       of insufficient strength, especially those using one of a few commonly
|       shared groups, may be susceptible to passive eavesdropping attacks.
|     Check results:
|       WEAK DH GROUP 1
|             Cipher Suite: TLS_DHE_RSA_WITH_AES_128_CBC_SHA
|             Modulus Type: Safe prime
|             Modulus Source: postfix builtin
|             Modulus Length: 1024
|             Generator Length: 8
|             Public Key Length: 1024
```

Figure 3.10

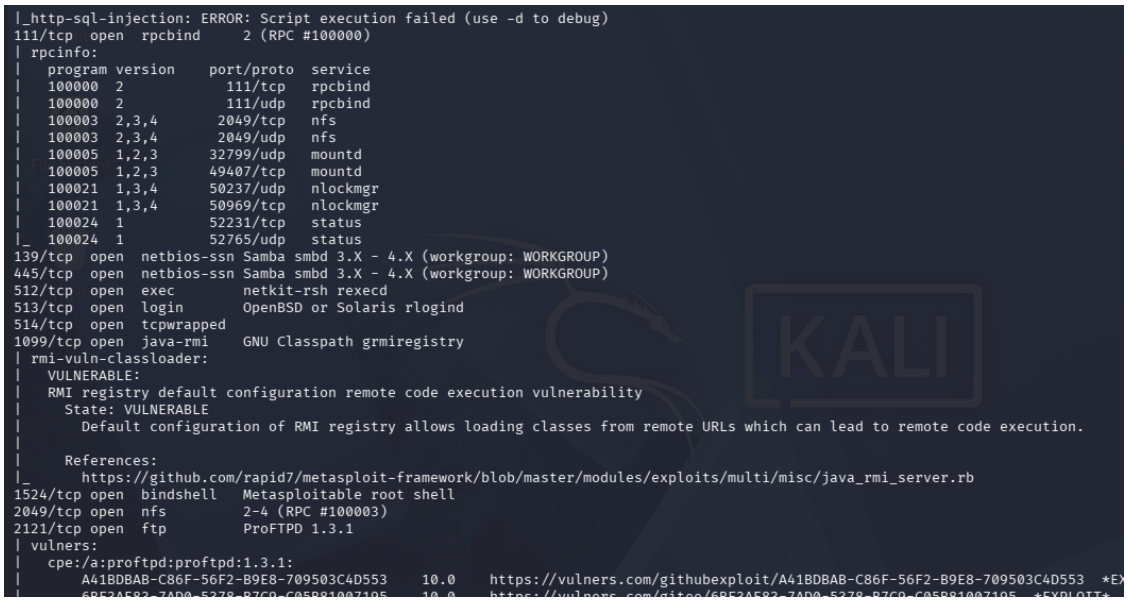


Figure 3.11

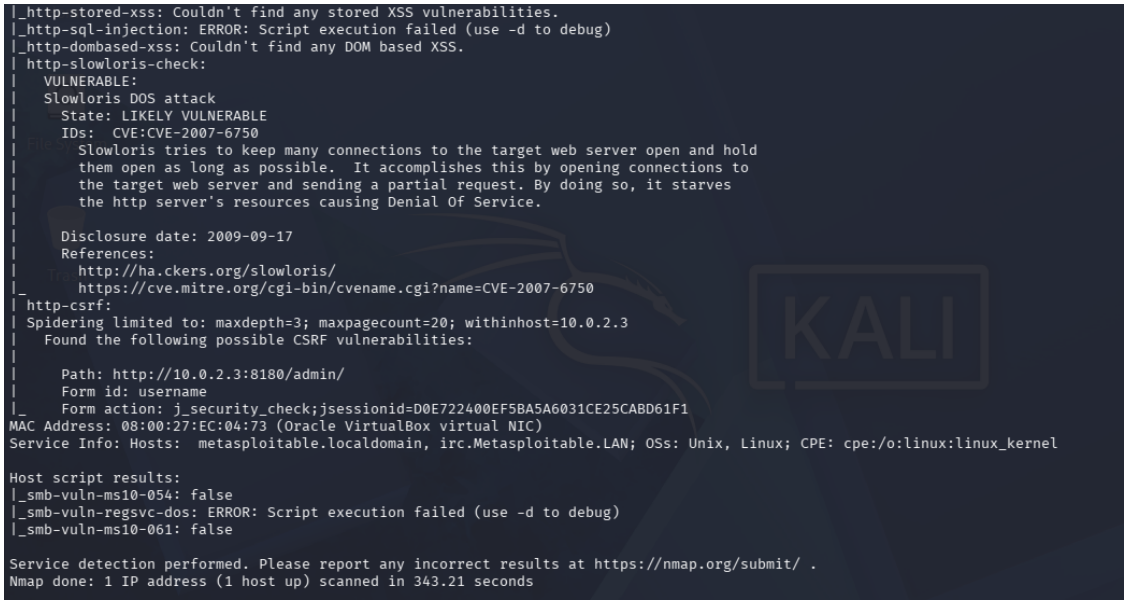


Figure 3.12



Figure 3.13

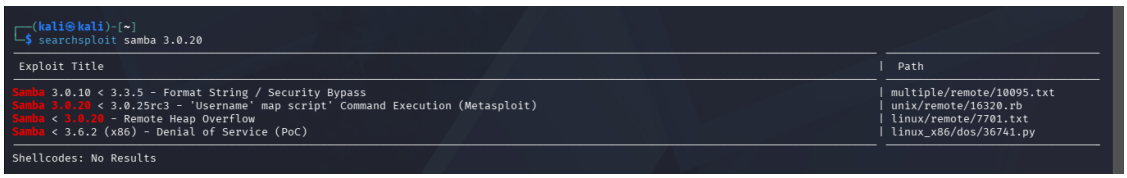


Figure 3.14



Figure 3.15

Task 3, Part 1 & 2: HTTP Enumeration and SSL Certificates

Figure 3.1 — HTTP directory enumeration using the http-enum NSE script

Task 3, Part 3: Vulnerability Scripts and Analysis

Figure 3.2 — Vulnerability scan (--script vuln) and Searchsploit cross-referencing

Interpretation: The NSE scans and Searchsploit mappings reveal direct, critical vulnerabilities matching the identified service versions — prominently the Samba 3.0.20 Remote Command Execution (CVE-2007-2447) and the vsFTPd 2.3.4 backdoor.

7. Task 4: Stealth & Firewall Evasion Techniques

Task 4 Screenshots

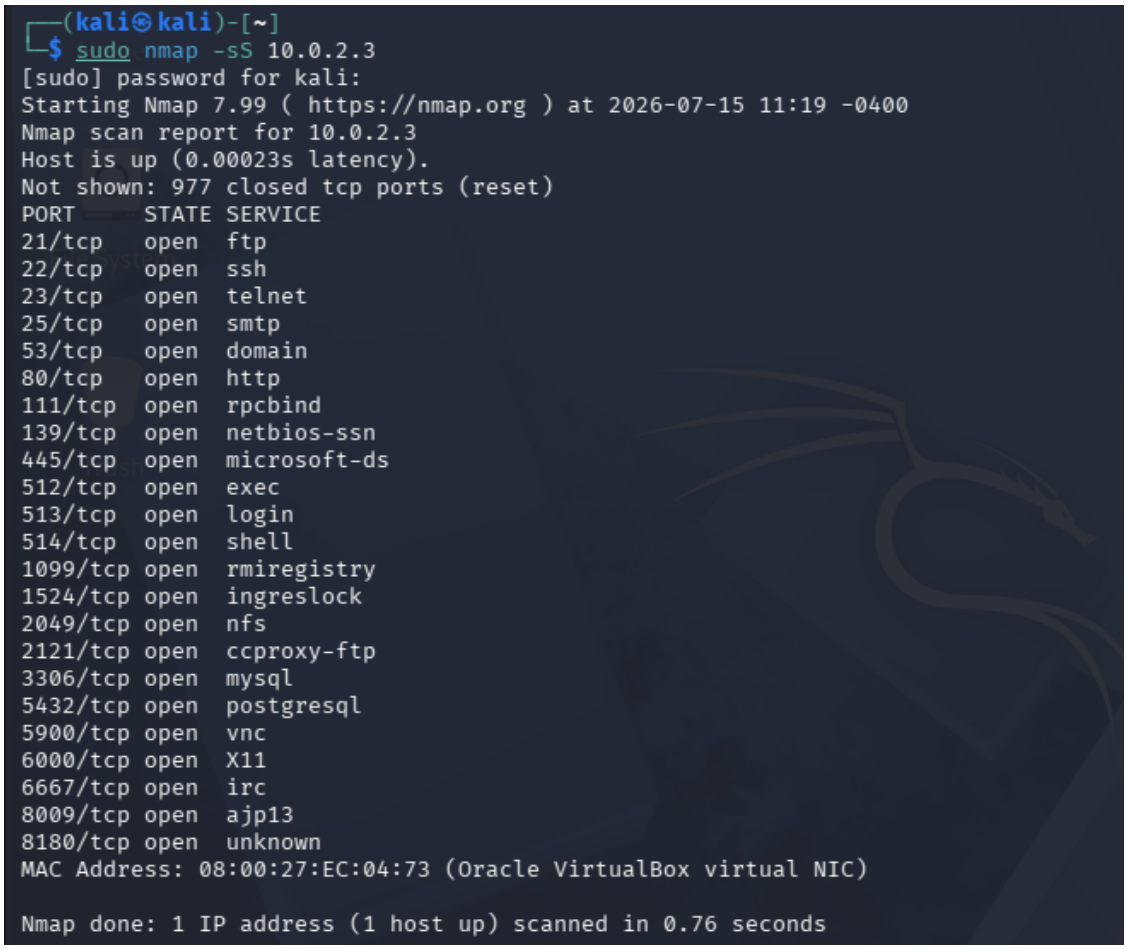

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS 10.0.2.3
[sudo] password for kali:
Starting Nmap 7.99 (https://nmap.org) at 2026-07-15 11:19 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00023s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.76 seconds

Figure 4.1

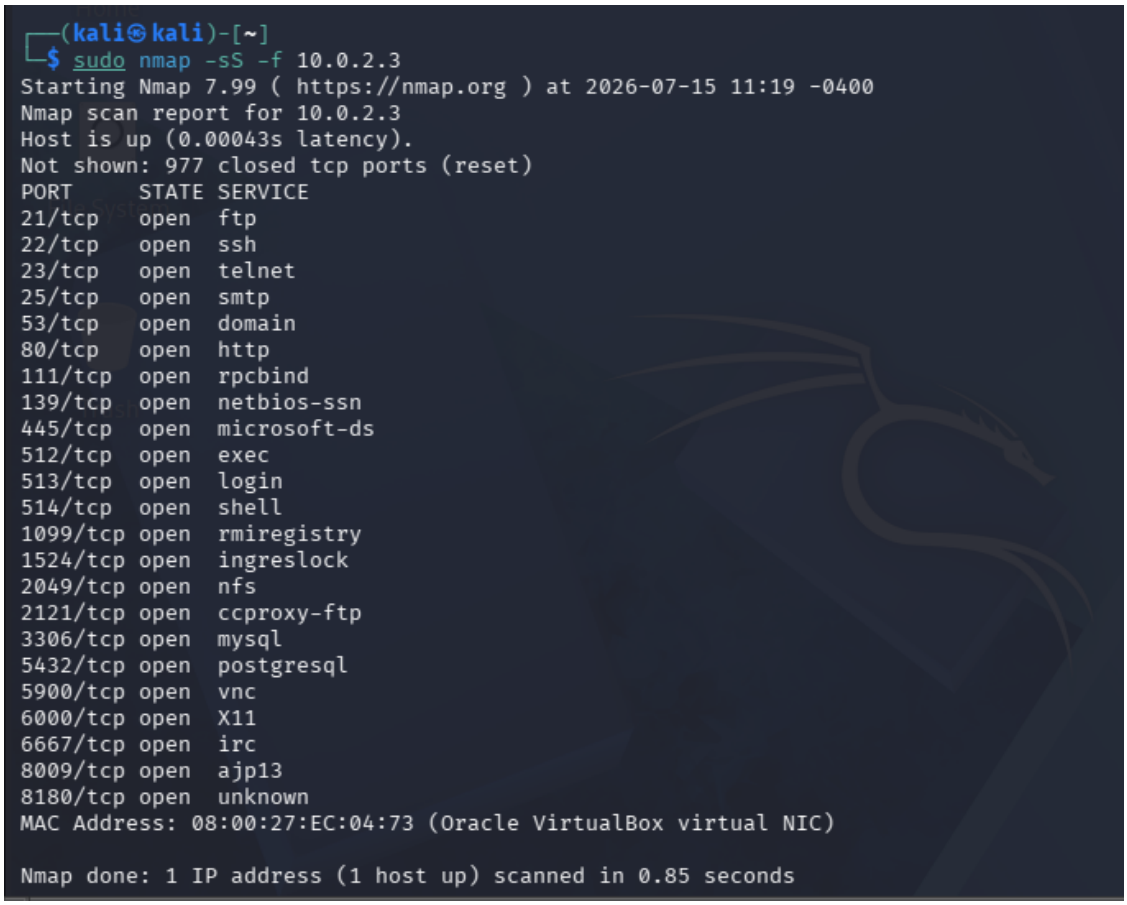

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -f 10.0.2.3
Starting Nmap 7.99 (https://nmap.org) at 2026-07-15 11:19 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00043s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.85 seconds

Figure 4.2

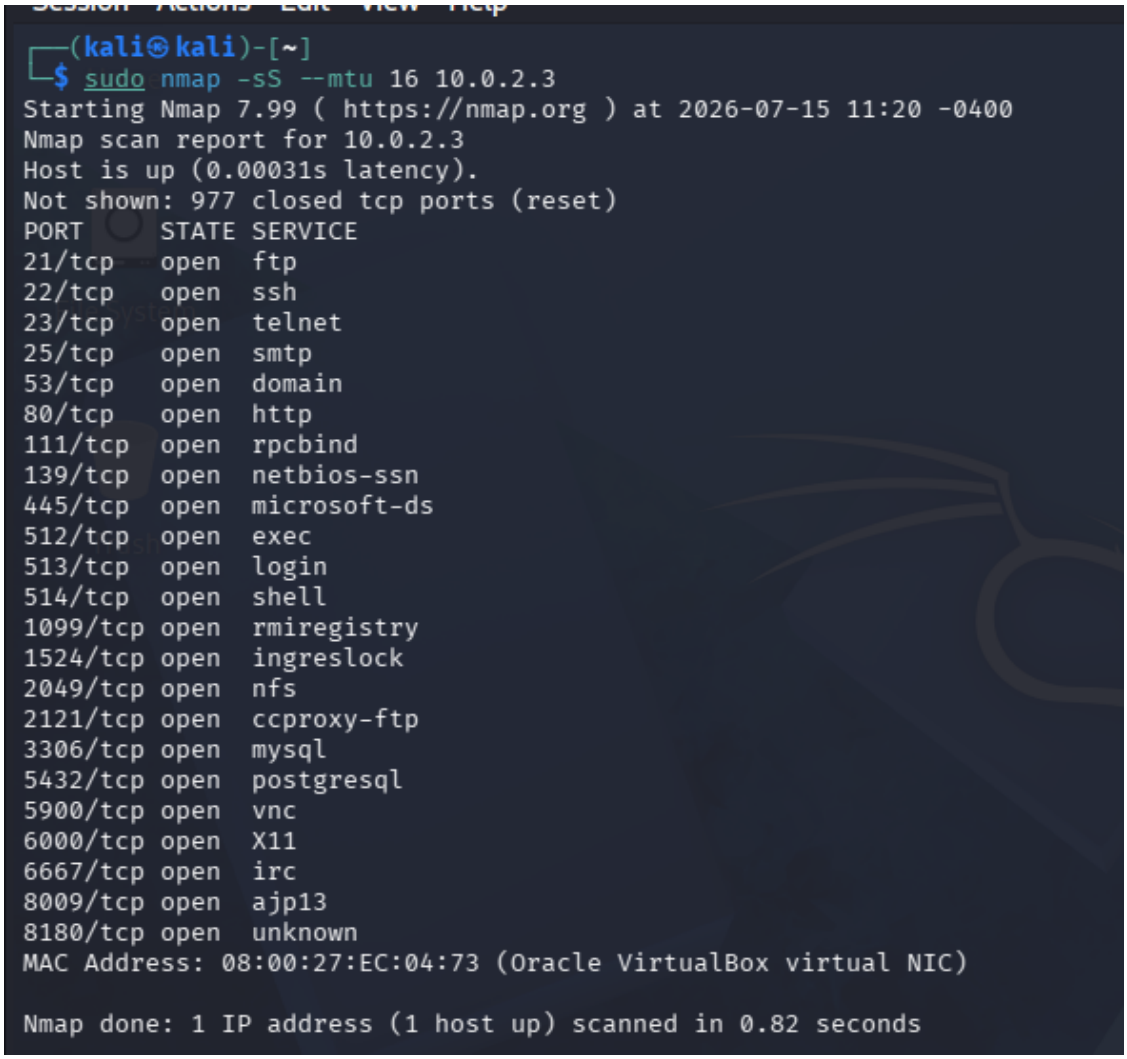

Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo nmap -sS --mtu 16 10.0.2.3
Starting Nmap 7.99 (https://nmap.org) at 2026-07-15 11:20 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00031s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.82 seconds

Figure 4.3

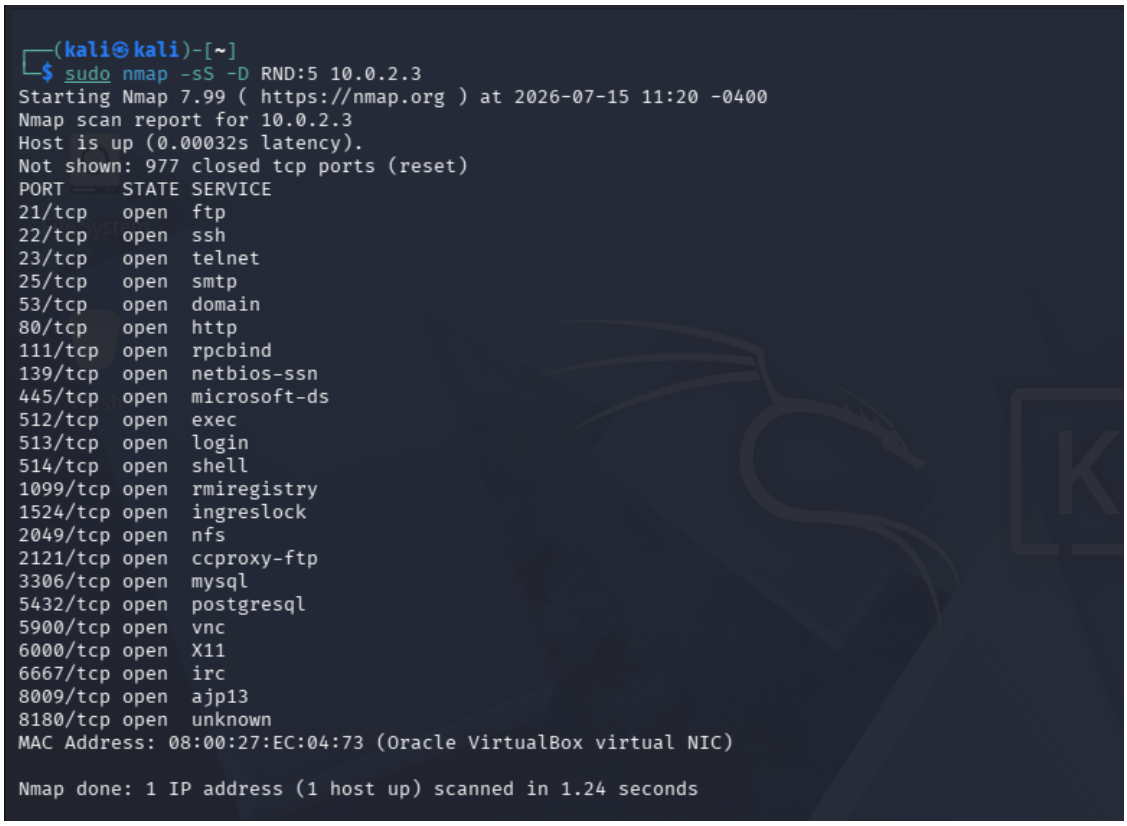


Figure 4.4

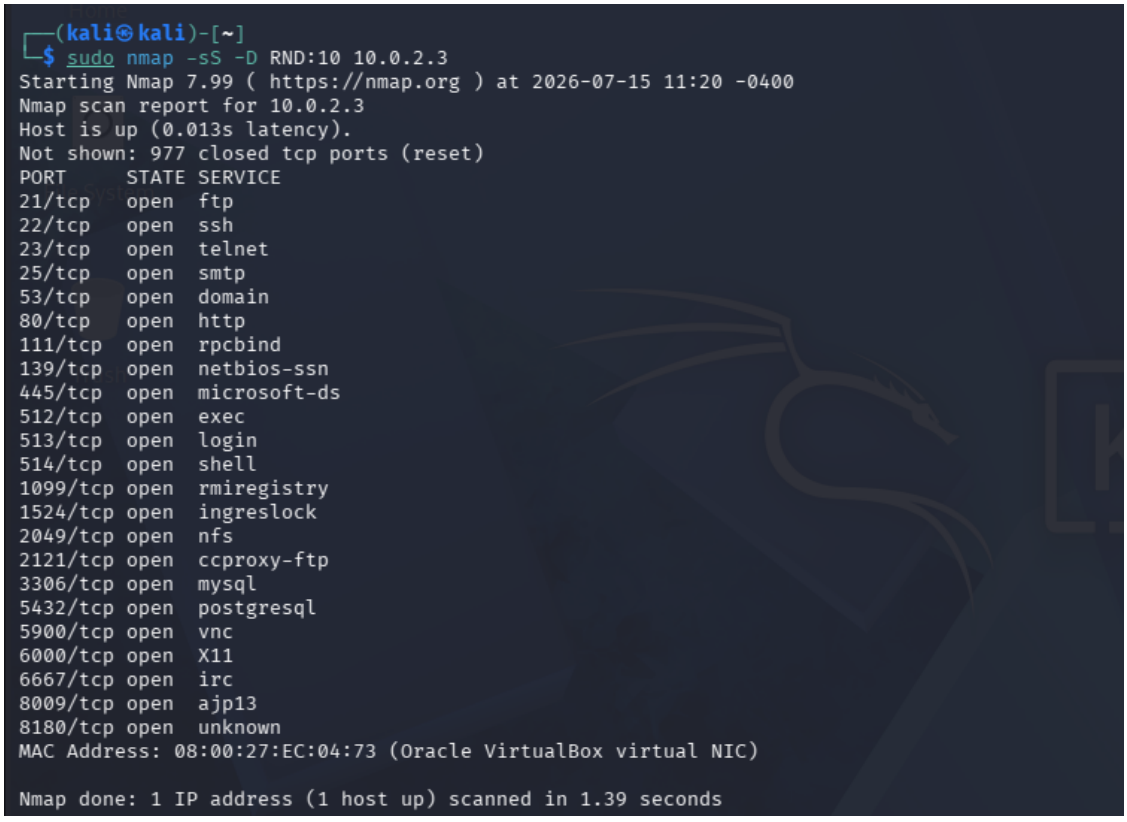


Figure 4.5

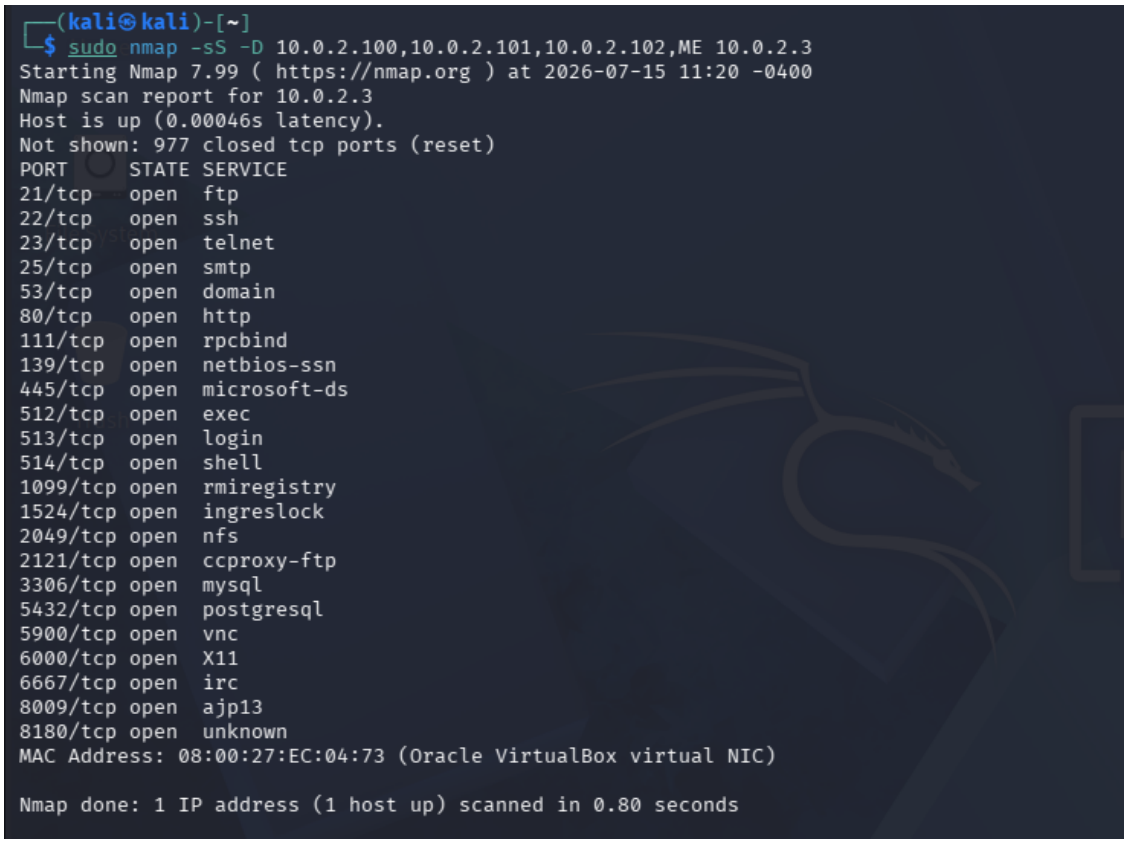

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -D 10.0.2.100,10.0.2.101,10.0.2.102,ME 10.0.2.3
Starting Nmap 7.99 (https://nmap.org) at 2026-07-15 11:20 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00046s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.80 seconds

Figure 4.6

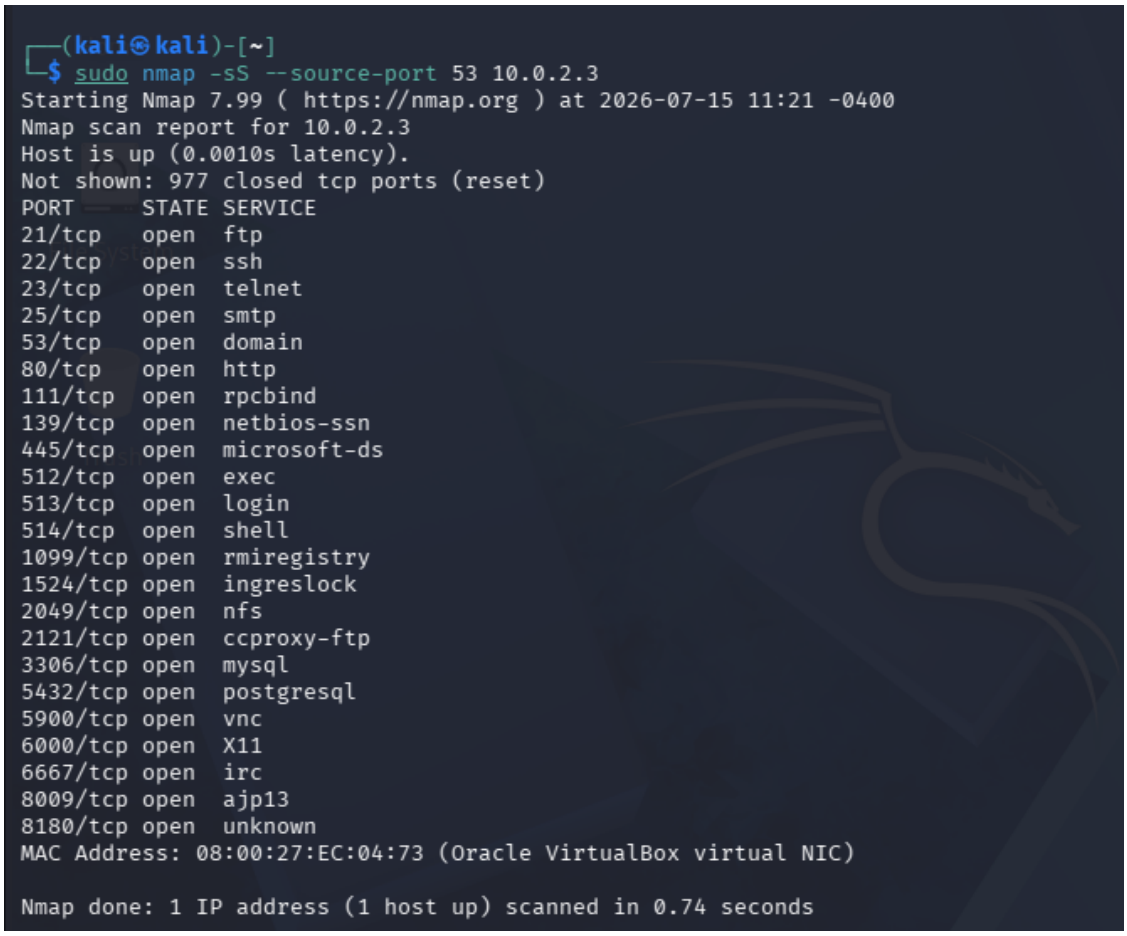

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS --source-port 53 10.0.2.3
Starting Nmap 7.99 (https://nmap.org) at 2026-07-15 11:21 -0400
Nmap scan report for 10.0.2.3
Host is up (0.0010s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.74 seconds

Figure 4.7

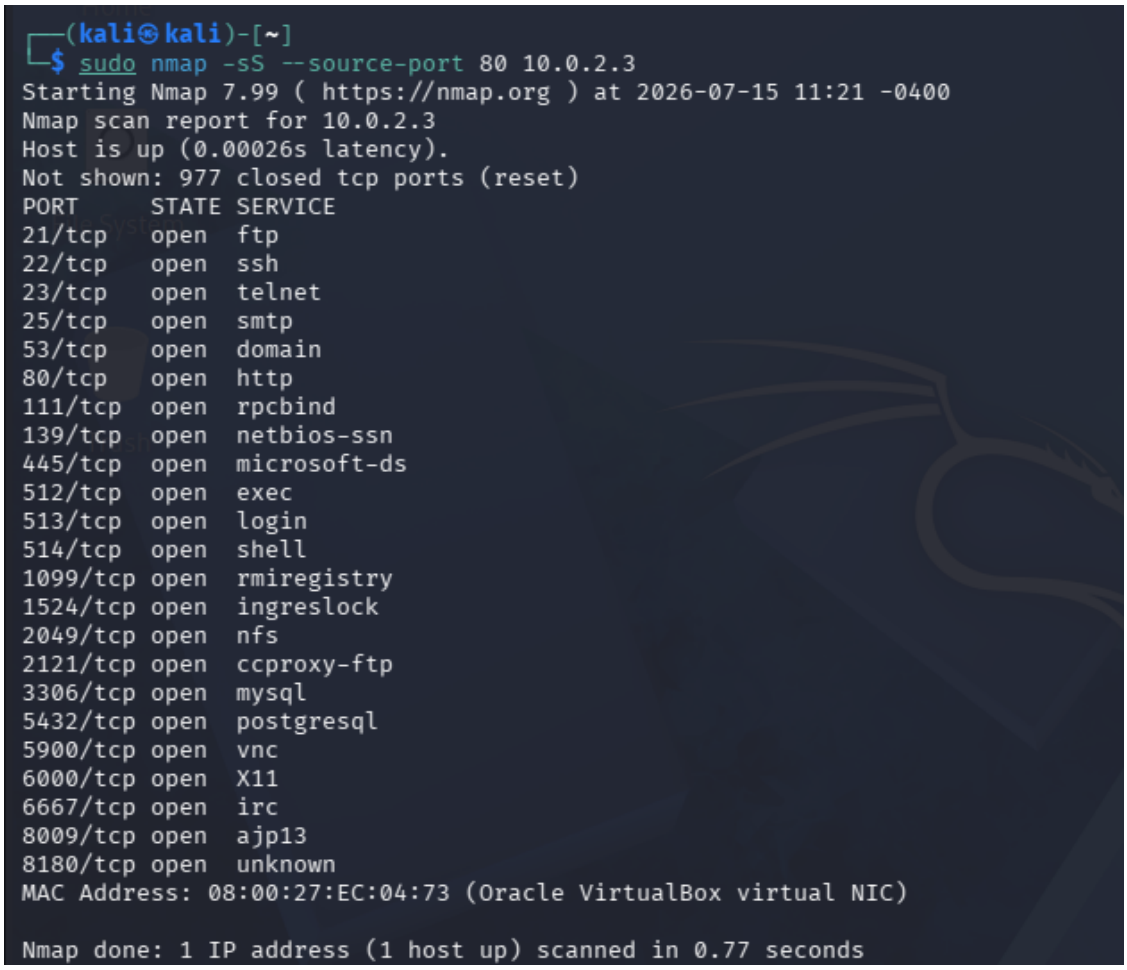

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS --source-port 80 10.0.2.3
Starting Nmap 7.99 (https://nmap.org) at 2026-07-15 11:21 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00026s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds

Figure 4.8

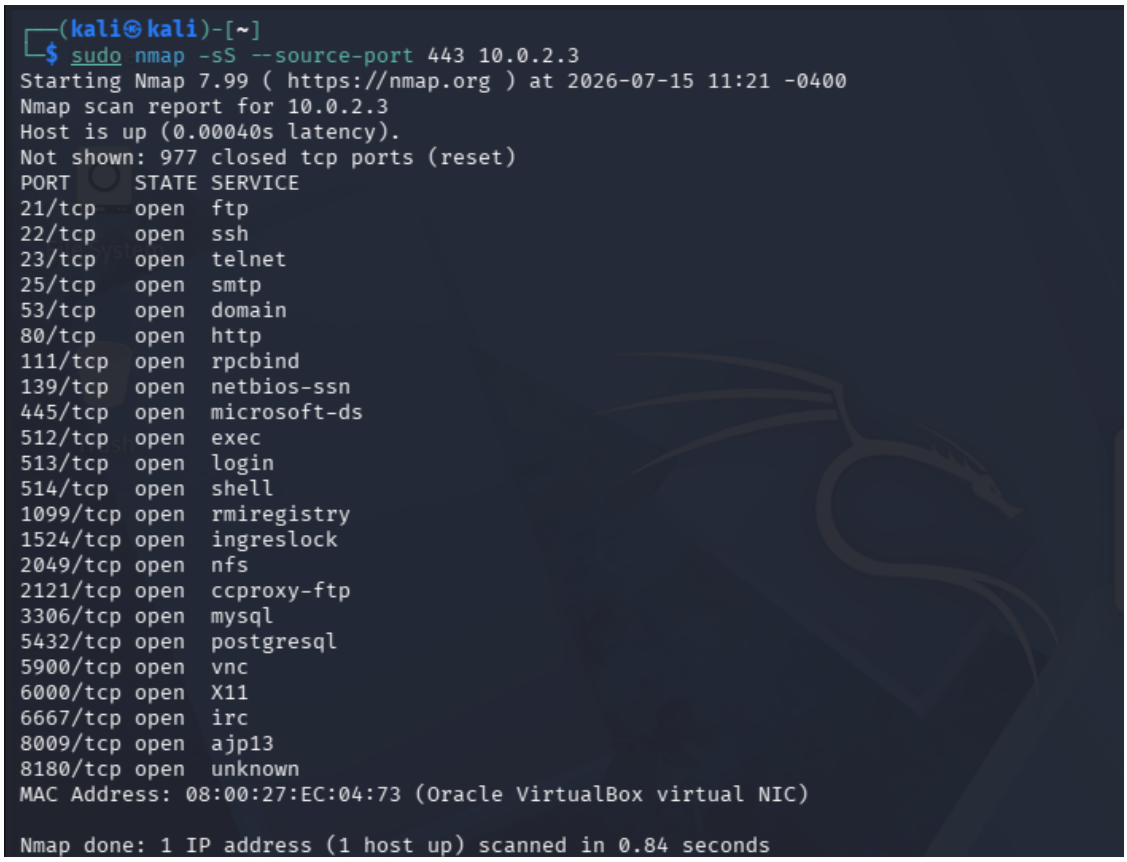

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS --source-port 443 10.0.2.3
Starting Nmap 7.99 (https://nmap.org) at 2026-07-15 11:21 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00040s latency).
Not shown: 977 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp open ftp
22/tcp open ssh
23/tcp open telnet
25/tcp open smtp
53/tcp open domain
80/tcp open http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1099/tcp open rmiregistry
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ccproxy-ftp
3306/tcp open mysql
5432/tcp open postgresql
5900/tcp open vnc
6000/tcp open X11
6667/tcp open irc
8009/tcp open ajp13
8180/tcp open unknown
MAC Address: 08:00:27:EC:04:73 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds

Figure 4.9

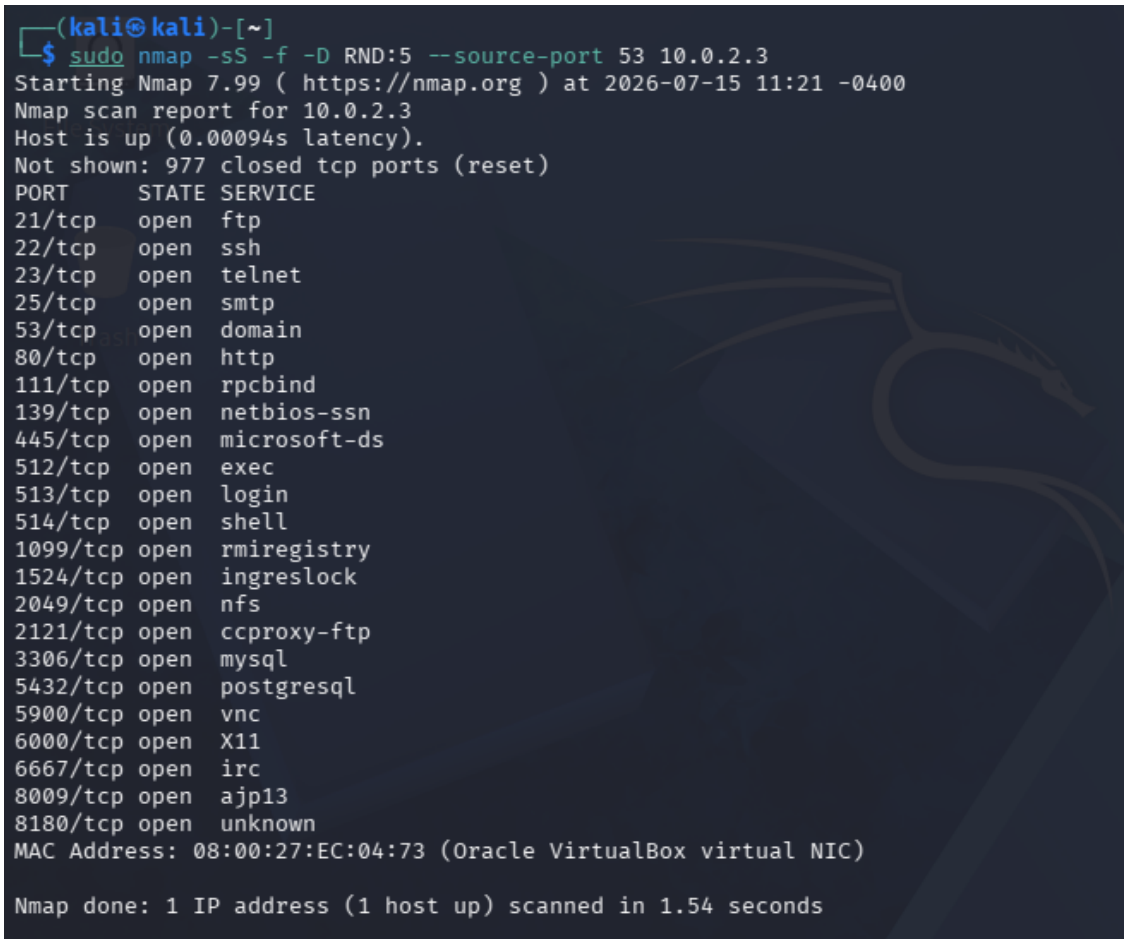


Figure 4.10

Task 4, Part 1: Packet Fragmentation and MTU Manipulation

Figure 4.1 — Packet fragmentation (-f) and custom MTU (--mtu 16) evasion scans

Task 4, Part 2: Decoys and Source Port Manipulation

Figure 4.2 — Decoy scanning (-D) and source port manipulation (--source-port 53)

Effectiveness: Decoys obscure the true source by mixing the scanner IP with randomly generated (RND:5) or specified IPs. Source port manipulation (--source-port 53) successfully bypasses rudimentary firewalls configured to trust incoming DNS traffic.

8. Challenges and Solutions

UDP Scanning Performance

Issue: Slow, stateless UDP scans. Solution: Addressed by utilizing --top-ports constraints and timing template tuning.

Service Version Ambiguity

Issue: Automated fingerprinting gaps. Solution: Resolved through manual Netcat and Telnet banner-grabbing sessions.

Vulnerability False Positives

Issue: NSE vulnerability scripts can over-report. Solution: Mitigated by mapping specific Nmap version outputs to trusted databases via Searchsploit.

Evasion Effectiveness

Issue: Verifying whether evasion techniques actually reduce visibility. Solution: Tested fragmentation, MTU scaling, and decoy IP mixing to map firewall handling characteristics.

9. Conclusions and Key Learnings

- Scan Type Matters: TCP Connect and SYN Stealth scans yield similar port results but carry vastly different network visibility footprints.
- Service Version is Critical: High-fidelity vulnerability discovery relies strictly on pinpointing exact software versions rather than generic port mapping.
- SMB is a High-Value Target: Extensive enumeration consistently points to unauthenticated SMB misconfigurations as a primary network foothold.

10. Defensive Strategies

Network Level

Deploy Deep Packet Inspection (DPI) to monitor for fragmented packets and implement rate limiting on connection attempts to thwart aggressive sweeps.

Service Level

Disable explicit banner versioning on exposed daemons (e.g., Apache, vsFTPd) and immediately apply critical patches for services like Samba.

11. Summary

Week 2 training progressed from passive reconnaissance to active network scanning and enumeration. Systematic use of Nmap scan types, service fingerprinting, NSE scripting, and evasion techniques revealed a substantial, exploitable attack surface on the Metasploitable 2 target — most notably the Samba usermap_script vulnerability and the vsFTPd 2.3.4 backdoor. These skills form the foundation for the vulnerability assessment and exploitation work planned for subsequent weeks.

12. Deliverables

Port	Protocol	Service	Version	Potential Vulnerabilities
21	TCP	FTP	vsftpd 2.3.4	CVE-2011-2523 (Backdoor)
22	TCP	SSH	OpenSSH 4.7p1	CVE-2008-1483, CVE-2008-1657
80	TCP	HTTP	Apache 2.2.8	CVE-2008-0005, CVE-2008-0456
445	TCP	SMB	Samba 3.0.20	CVE-2007-2444, CVE-2007-2447

Actionable Initial Access Point: Samba (Port 445). CVE-2007-2447 is exploitable via exploit/multi/samba/usermap_script in Metasploit, allowing immediate Remote Code Execution.